15.

Compound Interest

Type - 5

1. Mohit invested ₹10000 in two different schemes NSC and PPF at an annual compound interest rate of 14% and 11% respectively. If the total amount of interest received in 2 years is ₹ 2726, then what was the amount invested in PPF?

(a) ₹5000 (b) ₹4000
(c) ₹6000 (d) ₹7000

RRB RPF SI – 06/01/2019 (Shift-II)

Ans : (b) Let invested amount in NSC = ₹ x
Invested amount in PPF = ₹ (10000 – x)

$$x\left(1+\frac{14}{100}\right)^2+(10000-x)\left(1+\frac{11}{100}\right)^2=10000+2726$$

$$x\times\frac{114}{100}\times\frac{114}{100}+(10000-x)\frac{111}{100}\times\frac{111}{100}=12726$$

$$\frac{12996x}{10000}+(10000-x)\frac{12321}{10000}=12726$$

$$\frac{12996x}{10000}+\frac{123210000-12321x}{10000}=12726$$

$$12996x+123210000-12321x=127260000$$

$$675x=4050000$$

$$x=\frac{4050000}{675}=6000$$

Amount of PPF = 10000–6000 = ₹4000

2. Find the amount of interest for the third year at the rate of annual compound interest on ₹2000 at a rate of 40%.

(a) ₹1500 (b) ₹1600
(c) ₹1568 (d) ₹1750

RRB RPF Constable – 18/01/2019 (Shift-III)

Ans : (c) Principal = ₹2000
Rate = 40%

Interest for three year $=2000\left[\left(1+\frac{40}{100}\right)^3-1\right]$

$$=2000\times\left(\frac{7}{5}\right)^3-2000$$

$$=5488-2000=3488$$

Interest for two year $=2000\times\left(1+\frac{40}{100}\right)^2-2000$

$$=3920-2000=1920$$

Interest for third year = 3488 – 1920 = ₹1568

3. An amount of ₹16400 is borrowed and is to be repaid in 2 years in equal annual installments at the rate of 5% compound interest. Find the amount of annual payment.

(a) ₹ 7590 (b) ₹ 7495
(c) ₹ 7600 (d) ₹ 8820

RRB JE - 25/05/2019 (Shift-II)

Ans : (d) Given-

Total amount = installment $\left[\frac{1}{\left(1+\frac{r}{100}\right)^1}+\frac{1}{\left(1+\frac{r}{100}\right)^2}\right]$

$$16,400=x\left[\frac{1}{\left(1+\frac{5}{100}\right)^1}+\frac{1}{\left(1+\frac{5}{100}\right)^2}\right]$$

$$16,400=x\left[\frac{1}{\frac{21}{20}}+\frac{1}{\left(\frac{21}{20}\right)^2}\right]=x\left[\frac{20}{21}+\frac{400}{441}\right]$$

$$16,400=x\left[\frac{420+400}{441}\right]=x\times\frac{820}{441}$$

x = 20 × 441 = ₹ 8820

4. A women invested ₹200 at the beginning of every year at a 5% compound interest rate. At the end of the second year its total investment amount will be.

(a) ₹431 (b) ₹430.5
(c) ₹435 (d) ₹430

RRB RPF SI – 11/01/2019 (Shift-I)

Ans : (b) First Year, Principal = ₹200
Rate = 5% Time = 1 Year

Compound amount = Principal $\left(1+\frac{r}{100}\right)^t$

$$=200\left(1+\frac{5}{100}\right)^1=200\times\frac{21}{20}=₹\,210$$

Again for second year
Principal = ₹ 210 + ₹ 200 = ₹ 410
Rate = 5%, Time = 1 Year

New Compound amount = $410\left(1+\frac{5}{100}\right)^1=410\times\frac{21}{20}$

= ₹ 430.50

5. A certain sum of money was invested at compound interest, at 4% per annum for 2 years at compound interest, this amount becomes ₹338. Find the amount.

(a) ₹320 (b) ₹ 312.5
(c) ₹318.53 (d) ₹315

RRB JE - 25/05/2019 (Shift-III)

Ans : (b) Given-

P = ?, r = 4%, Amount = ₹338, t = 2 Year

As per the question-

$$\text{Amount} = P\left(1+\frac{r}{100}\right)^t$$

$$338 = P\left(1+\frac{4}{100}\right)^2$$

$$338 = P\times\left(\frac{26}{25}\right)^2$$

$$P = \frac{211250}{676}$$

P = ₹312.5

6. An amount is invested for two years at 20% compound interest. If the interest is payable on a half yearly basis, then it receives ₹482 more than that which was payable on an annual basis. Find the amount.

(a) ₹ 19500 (b) ₹ 20000
(c) ₹ 21800 (d) ₹ 30000

RRB JE - 26/06/2019 (Shift-I)

Ans. (b) Let Principal amount = ₹x

Rate of Compound interest = 20%

$$\text{Compound interest} = \text{Principal}\left[\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}-1\right]$$

$$= x\left[\left(1+\frac{20}{100}\right)^2-1\right]$$

$$= x\left[\left(1+\frac{1}{5}\right)^2-1\right]$$

$$= x\left[\left(\frac{6}{5}\right)^2-1\right]$$

$$= x\left[\frac{36}{25}-1\right]$$

$$= \frac{11x}{25}$$

If interest are half yearly payable

Rate of interest = 10%

Time = 4 Year

$$\text{Compound interest} = \text{Principal}\left[\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}-1\right]$$

$$= x\left[\left(1+\frac{10}{100}\right)^4-1\right]$$

$$= x\left[\left(\frac{11}{10}\right)^4-1\right]$$

$$= x\left[\frac{11\times11\times11\times11-10000}{10000}\right]$$

$$= x\left[\frac{14641-10000}{10000}\right] = \frac{4641x}{10000}$$

As per the question,

$$\frac{4641x}{10000}-\frac{11x}{25} = 482$$

$$\frac{4641x-4400x}{10000} = 482$$

$$\frac{241x}{10000} = 482$$

x = 2 × 10000 = ₹20000

7. A fixed amount becomes ₹2420 in two years at a rate of compound interest and ₹2662 in three years, the interest was calculated on an annual compound basis. Find the amount and rate of interest annually.

(a) ₹1000 and 12% (b) ₹ 2000 and 10%
(c) ₹ 2250 and 15% (d) ₹ 2500 and 5%

RRB JE - 31/05/2019 (Shift-II)

Ans : (b) Let that amount P and interest rate is r%

$$\text{Compound amount} = P\left(1+\frac{r}{100}\right)^n$$

According to the question-

$$2420 = P\left(1+\frac{r}{100}\right)^2 \quad \text{----(i)}$$

$$2662 = P\left(1+\frac{r}{100}\right)^3 \quad \text{----(ii)}$$

Dividing equation (ii) in (i)

$$1+\frac{r}{100} = \frac{2662}{2420}$$

r = 10%

from equation (i) - $2420 = P\left(1+\frac{10}{100}\right)^2$

$$P = 2420\times\frac{10}{11}\times\frac{10}{11} = ₹2000$$

8. Mr. Marthi invested ₹16000 in a scheme. How much money will be get when he becomes on adult if he invests it for 9 months at a compound interest rate of 20% per annum.

(a) ₹ 18,523 (b) ₹ 18,521
(c) ₹ 18,524 (d) ₹ 18,522

RRB RPF Constable – 17/01/2019 (Shift-I)

Ans : (d)

Compound interest Rate on quarter $= \frac{20}{4}\% = 5\%$

time = 9 months = 3 quarter

Received amount on being adult of marthy

$$= 16000\left[1+\frac{5}{100}\right]^3$$

$$\left\{\because A = P\left(1+\frac{r}{100}\right)^n\right\}$$

$$= 16000\left(\frac{21}{20}\right)^3 = \frac{16000\times21\times21\times21}{20\times20\times20} = ₹18522$$

9. A person borrowed money at 9% simple interest and invested it at 10% compound interest for 3 years After 3 years he received profit of ₹1952. How much money did he borrow?

(a) ₹ 30000 (b) ₹ 32000
(c) ₹ 33000 (d) ₹ 32543

RRB Group-D – 06/12/2018 (Shift-III)

Ans. (b) : Suppose borrowed amount = ₹ x

Compound interest = total Amount – Principal

Profit = Compound interest – simple interest

As per the question,

$$x\left[\left(1+\frac{10}{100}\right)^3-1\right]-\left[\frac{x\times9\times3}{100}\right]=1952$$

$$\Rightarrow x\left[\left(\frac{11}{10}\right)^3-1\right]-\left[\frac{x\times9\times3}{100}\right]=1952$$

$$\Rightarrow x\left[\frac{1331}{1000}-1\right]-\left[\frac{x\times9\times3}{100}\right]=1952$$

$$\Rightarrow x\left[\frac{1331-1000}{1000}\right]-\left[\frac{x\times9\times3}{100}\right]=1952$$

$$\Rightarrow x\left[\frac{331}{1000}\right]-\frac{27x}{100}=1952$$

$$\Rightarrow \frac{x(331-270)}{1000}=1952$$

$$\Rightarrow \frac{x\times61}{1000}=1952$$

$$\Rightarrow x = 32\times1000 = ₹32000$$

10. The money invested for two years is to be compounded on an annual basis, at a rate of 20% per annum. At maturity it becomes ₹324. What was the initial amount invested.

(a) ₹240 (b) ₹200
(c) ₹250 (d) ₹225

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (d) Let Amount = ₹ P

time = 2 Year

, Rate = 20% Annual

$$324 = P\left(1+\frac{20}{100}\right)^2$$

$$324 = P\times\frac{6}{5}\times\frac{6}{5}$$

$$P = \frac{324\times25}{36}$$

P = ₹225

11. When there is an annual compound interest at the rate of around 9% per annum, what will be the amount of ₹5000 after 2 years.

(a) ₹5,940 (b) ₹9,950
(c) ₹5,970 (d) ₹5,936

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (a)

Principal (P) = ₹ 5000

Time (t) = 2

Rate (r) = 9%

$$\text{Compound amount} = \text{Principal}\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}$$

$$= 5000\left(1+\frac{9}{100}\right)^2$$

$$= 5000\times\frac{109}{100}\times\frac{109}{100}$$

$$= \frac{5\times109\times109}{10} = 5940.5 \approx ₹\ 5940$$

12. Shyam deposits ₹x for 2 years at 8% per annum compound annual interest, which is ₹72900. Then what is the value of x?

(a) ₹60,500 (b) ₹62,000
(c) ₹60,000 (d) ₹62,500

RRB Group-D – 04/10/2018 (Shift-II)

Ans : (d) r = 8%

n = 2 Year

A = ₹ 72,900

$$A = P\left(1+\frac{R}{100}\right)^n,\quad 72900 = P\left[1+\frac{8}{100}\right]^2$$

$$72900 = P\times\frac{108}{100}\times\frac{108}{100}$$

$$P = \frac{72900\times100\times100}{108\times108}$$

P = ₹ 62500

13. If ₹10000 is invested at the rate of 20% per annum, at half yearly compound interest, then what will be the amount after 18 months.

(a) ₹16000 (b) ₹ 13310
(c) ₹13000 (d) ₹ 13140

RRB Paramedical Exam – 20/07/2018 (Shift-I)

Ans : (b) time (t) = 18 month = 3 Half yearly

Rate (r) = 20% Annual = 10% Half yearly

Then required Amount = Principal $\left(1+\frac{r}{100}\right)^t$

$$=10000\left(1+\frac{10}{100}\right)^3$$

$$=10000\times\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10}=₹13310$$

14. If you have invested ₹100 on a 40% annual interest, which would be a quarterly compounded. How much amount will you get after one year?

(a) ₹148.11 (b) ₹146.41
(c) ₹145.61 (d) ₹149.01

RRB Group-D – 04/10/2018 (Shift-I)

Ans. (b) interest Rate = 40% Annual

$\therefore$ quarter interest Rate $=\frac{40\times3}{12}=10\%$

time = 4 quarter

Amount after 1 year $=100\times\left(1+\frac{10}{100}\right)^4$

$$=100\times\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10}$$

$$=\frac{121\times121}{100}$$

$$=₹146.41$$

15. A women invests ₹2000 per year at the beginning of every year at a compound interest rate of 5%. How much will she invest at the end of the second year.

(a) ₹ 4305 (b) ₹ 430
(c) ₹ 4355 (d) ₹ 4350

RRB NTPC 17.01.2017 Shift-1

Ans : (a) Received amount in first year

= invested amount + simple interest

$$=2000+\frac{2000\times5\times1}{100}=2100$$

Received amount in second year

= invested amount + simple interest

$$=(2100+2000)+\frac{4100\times5\times1}{100}=4100+205=4305$$

Hence the women invest ₹4305 at the end of second year.

16. A person borrows a certain amount from a bank for 3 years at a rate of 7% compound interest annually. If he paid ₹85966 as total interest, then what was the amount borrowed?

(a) ₹462,000 (b) ₹382,000
(c) ₹354,000 (d) ₹428,000

RRB NTPC 04.04.2016 Shift : 1

Ans : (b) Let borrowed amuont = ₹x

Amount (A) = Principal + interest

= x + 85966

$$\because A=P\left(1+\frac{r}{100}\right)^n$$

$$x+85966=x\left(1+\frac{7}{100}\right)^3$$

$x + 85966 = x\,(1.07)^3$

$x + 85966 = 1.225043x$

$0.225043x = 85966$

$$x=\frac{85966}{0.225043}$$

$x = 381998.107$, $x = ₹382000$

17. After two years of ₹80000 from compound interest compounded annually, what will be the amount at the annual rate of 20%.

(a) ₹97,240 (b) ₹117,128
(c) ₹115,200 (d) ₹120,000

RRB NTPC 02.04.2016 Shift : 1

Ans : (b) $\because$ interest is half yearly

$\therefore$ Time = 2 Year = 4 Half year

Rate $=\frac{20}{2}=10\%$ Half yearly

$$\therefore A=80000\left(1+\frac{10}{100}\right)^4$$

$$=80000\times\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10}=₹117,128$$

18. A takes a fixed amount from a bank at a rate of 8% interest with interest compound half yearly. If he paid ₹196851 after one and a half year, find the principal.

(a) ₹ 168,000 (b) ₹ 175,000
(c) ₹ 179,000 (d) ₹ 184,000

RRB NTPC 31.03.2016 Shift : 3

Ans : (b) Given- Rate (r) = 8% Annual, = 4% Half yearly

Time n = 1 Year = 2 Half yearly

1.5 Year = 3 Half yearly

$$\text{Principal}=\frac{196851}{\left(1+\frac{4}{100}\right)^3}=\frac{196851}{(26/25)^3}$$

$$=\frac{196851\times25\times25\times25}{26\times26\times26}$$

$$=174999.82=₹175000$$

19. If 'P' expresses the principal invested at the semi annual rate for a period of 2 years at a 5% interest rate, then after 2 years how much amount will he receive?

(a) $P(1.025)^4$ (b) $P(1.1)$
(c) $P(1.05)^2$ (d) $P(1.025)^2$

RRB NTPC 18.01.2017 Shift : 2

Ans : (a) 2 year = 4 Half year

$$R = \left(\frac{5}{2}\right)\% \text{ Half year}$$

$$A = P\left(1+\frac{R}{100}\right)^n$$

$$A = P\left(1+\frac{5}{2\times100}\right)^4$$

$$A = P\left(\frac{41}{40}\right)^4$$

$$A = P(1.025)^4$$

20. A fixed amount becomes ₹7200 in 2 years at a compound interest rate of 20% per annum. Find the principal amount.

(a) ₹4800 (b) ₹6000
(c) ₹5400 (d) ₹5000

RRB NTPC 18.01.2017 Shift : 3

Ans : (d) Compound amount $= P\left(1+\frac{r}{100}\right)^n$

$$7200 = P\left(1+\frac{20}{100}\right)^2$$

$$7200 = P\left(1+\frac{20}{100}\right)^2$$

$$7200 = P\times\frac{36}{25}$$

P = 200 × 25 = ₹5000

21. Prakash invests in a FD. What will be the matured amount with an annual compound interest rate of 20% for 6 months at ₹13000, interest is compounded quarterly?

(a) ₹14332.25 (b) ₹14332.5
(c) ₹14332.75 (d) ₹14332

RRB NTPC 07.04.2016 Shift : 3

Ans :(b) P = ₹13000

T = 6 months $= \frac{6}{3}$ quarter = 2 quarter

$R = \frac{20}{4}\% = 5\%$ quarterly

$$A = P\left(1+\frac{R}{100}\right)^2 = 13000\times\left(1+\frac{5}{100}\right)^2$$

Compound amount

$$= 13000\times\left(1+\frac{1}{20}\right)^2 = 13000\times\left(\frac{21}{20}\right)^2$$

$$= 13000\times\frac{21}{20}\times\frac{21}{20} = ₹14332.5$$

22. Mr. Akhil deposited ₹13500 in a fixed deposit find the total money for 6 months at the rate of compound interest of 20% per annum if the interest is compound for every 3 months.

(a) ₹ 14, 883.35 (b) ₹ 14,883.75
(c) ₹ 14,883.5 (d) ₹ 14,883

RRB NTPC 19.01.2017 Shift : 1

Ans : (b) When Rate is quarter $= \frac{20}{4} = 5\%$

time = 6 month = 2 quarter

As per the question,

$$= 13500\left[1+\frac{5}{100}\right]^2$$

$$= 13500\times\frac{21\times21}{20\times20} = ₹14883.75$$

23. A women invested ₹4000 at the beginning of the year at a compound interest rate of 5% per annum. What will be his invested amount at the end of the second year?

(a) ₹ 4410 (b) ₹ 4615
(c) ₹ 5000 (d) ₹ 4010

RRB NTPC 19.01.2017 Shift : 3

Ans : (a) $S.I = \frac{PRT}{100} = \frac{4000\times5\times1}{100} = 200$

$$C.I = \frac{4200\times105}{100} = ₹4410$$

24. Rakesh invests ₹10000 in a fixed deposit scheme for 2 years at a 5% annual rate of compound interest. How much amount will he get on maturity of the fixed deposit?

(a) ₹ 10000 (b) ₹ 8500
(c) ₹ 11050 (d) ₹ 11025

RRB NTPC 22.04.2016 Shift : 1

Ans : (d) Given-

Principal (P) = ₹10000

Annual Rate (R) = 5%

time (t) = 2 Year

Formula– Compound Amount (A) $= P\left(1+\frac{R}{100}\right)^t$ from-

$$A = 10000\left(1+\frac{5}{100}\right)^2$$

$$= 10000\left(\frac{21}{20}\right)^2$$

$$= 10000\times\frac{441}{400}$$

$$= 25\times441 = ₹11025$$

25. A person invests ₹20000 for 1 year at an annual interest rate of 15%. If the interest is compounded half yearly, then the amount received by him at the end of the year will be.

(a) ₹16537.50 (b) ₹23112.50
(c) ₹18112.75 (d) ₹22175.30

RRB NTPC 22.04.2016 Shift : 3

Ans : (b) Principal (P) = ₹20,000
Time (n) = 2 (Half yearly)
Rate $(R) = \frac{15}{2}\%$ (Half yearly)

$$A = P\left(1+\frac{r}{100}\right)^n$$

$$= 20000\left(1+\frac{15}{2\times100}\right)^2$$

$$= 20000\times\frac{43}{40}\times\frac{43}{40} = ₹23112.50$$

26. Calculate the principal if an amount of ₹ 441 is received on compound interest at the rate of 5% per annum after 2 years
(a) ₹400 (b) ₹390
(c) ₹380 (d) ₹350
RRB NTPC 27.04.2016 Shift : 3

Ans : (a) As per the question,

$$\text{Amount} = \text{Principal}\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}$$

$$441 = \text{Principal}\left(1+\frac{5}{100}\right)^2$$

$$441 = \text{Principal}\left(\frac{21}{20}\right)^2$$

$$441 = \text{Principal}\times\frac{441}{400}$$

$$\text{Principal} = \frac{441\times400}{441} = ₹400$$

27. Gitesh took a loan for 4 years at 5% compound interest. If the total interest paid was ₹ 431.01, Calculate the principal.
(a) ₹2000 (b) ₹2050
(c) ₹2100 (d) ₹2150
RRB NTPC 27.04.2016 Shift : 3

Ans : (a)

$$\text{Compound interest} = \text{Principal}\left[\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}-1\right]$$

$$431.01 = \text{Principal}\left[\left(1+\frac{5}{100}\right)^4-1\right]$$

$$431.01 = \text{Principal}\left[\frac{21}{20}\times\frac{21}{20}\times\frac{21}{20}\times\frac{21}{20}-1\right]$$

$$431.01 = \text{Principal}\left[\frac{194481}{160000}-1\right]$$

$$431.01 = \text{Principal}\left[\frac{194481-160000}{160000}\right]$$

$$431.01 = \text{Principal}\times\frac{34481}{160000}$$

$$\text{Principal} = \frac{431.01\times160000}{34481} = 1999.99 = ₹\,2000$$

28. A women deposited ₹100 at an annual interest rate of 10%. The women received ₹121 at the end of the period at an annual rate of compound interest. How long did he keep the money in the bank?
(a) 1 Year (b) 2 Year
(c) 2.5 Year (d) 3 Year
RRB NTPC 28.04.2016 Shift : 3

Ans : (b) Given–
Principal (P) = ₹100
Amount (A) = ₹121
Rate (R) = 10%
time t = ?

Formula– $A = P\left(1+\frac{R}{100}\right)^t$ से–

$$121 = 100\left(1+\frac{10}{100}\right)^t$$

$$\frac{121}{100} = \left(\frac{11}{10}\right)^t$$

$$\left(\frac{11}{10}\right)^2 = \left(\frac{11}{10}\right)^t$$

t = 2 Year

29. If a certain amount becomes ₹6655 at a compound interest rate of 10% in 3 years. Find the amount.
(a) ₹5000 (b) ₹5500
(c) ₹4500 (d) ₹4800
RRB NTPC 30.04.2016 Shift : 1

Ans : (a) $A = P\left(1+\frac{r}{100}\right)^n$

$$6655 = P\left(1+\frac{10}{100}\right)^3$$

$$6655 = \frac{1331P}{1000}$$

$$P = \frac{1000\times6655}{1331} = ₹5000$$

30. Rakesh invested ₹10000 in two different schemes in NSC and PPF at an annual compound interest rate of 14% and 11% respectively. If the total interest for 2 years is ₹2726, then what was the amount invested in it (NSC)?
(a) ₹5000 (b) ₹4000
(c) ₹6000 (d) ₹7000
RRB NTPC 30.04.2016 Shift : 2

Ans : (c) Let invested amount in NSC is ₹x. then invested amount in PPF will be ₹ (10000-x).
Total Amount = 10,000 + 2726 = ₹12726
As per the question,

$$12726 = x\left(1+\frac{14}{100}\right)^2+(10,000-x)\times\left(1+\frac{11}{100}\right)^2$$

$$12,726 = x \times \frac{114\times114}{100\times100} + 10,000 \times \frac{111\times111}{100\times100} - x \times \frac{111\times111}{100\times100}$$

$\Rightarrow 12,726\times10,000 = 12,996x + 12,321\times10,000 - 12,321x$

$\Rightarrow (12,726\times10,000 - 12321\times10,000) = 12996x - 12321x$

$10,000(405) = 675x$

$$x = \frac{40,50,000}{675} = ₹6000$$

31. If ₹ 2,000 is invested at the rate of 20% per annum, compounded half-yearly, then the amount after 18 months will be:

(a) ₹2,628 (b) ₹2,662
(c) ₹3,200 (d) ₹2,600

RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (b) Principal (P) = ₹2000

Rate (r) = 20% Annual, half yearly interest is payable

so Rate (r) = $\frac{20}{2}$, = 10%, time (T) = 18 month = 3 Half year

Formula- Compound interest

$$= \text{Principal}\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}} = 2000\left(1+\frac{10}{100}\right)^3$$

$$= 2000\left(\frac{11}{10}\right)^3 = 2\times1331 = 2662$$

After 18 month amount will be ₹2662

Type - 2

32. In how many years ₹1728 will become ₹2197 at $8\frac{1}{3}$ annual rate of compound interest.

(a) 2 Year (b) $1\frac{1}{2}$ Year
(c) 3 Year (d) $2\frac{1}{2}$ Year

RRB RPF SI – 16/01/2019 (Shift-III)

Ans : (c) Principal = ₹1728

Amount = ₹ 2197

Rate (R) = $8\frac{1}{3}$

Time (n) = ?

$$\text{Amount} = \text{Principal}\left(1+\frac{R}{100}\right)^n$$

$$2197 = 1728\left(1+\frac{8\frac{1}{3}}{100}\right)^n = 1728\left(1+\frac{25}{300}\right)^n$$

$$\frac{2197}{1728} = \left(\frac{13}{12}\right)^n$$

$$\left(\frac{13}{12}\right)^3 = \left(\frac{13}{12}\right)^n$$

n = 3 Year

33. If the amount of ₹64 increases to ₹83.20 in 2 years, then at the same percentage rate of simple interest, how much amount of ₹86 will become in 4 years

(a) ₹ 114.8 (b) ₹ 124.70
(c) ₹ 127.40 (d) ₹ 137.60

RRB JE - 22/05/2019 (Shift-III)

Ans : (d) Let Rate is R%

$$\text{simple interest} = \frac{P\times R\times T}{100}$$

simple interest = Amount − Principal

= 83.20 – 64 = 19.20

$$\text{Simple interest} = \frac{P\times R\times T}{100}$$

$$19.200 = \frac{64\times R\times 2}{100}, \quad \frac{1920}{128} = R$$

R = 15%

Rate = 15%, Principal = ₹86, time = 4 Year,

Amount = ?

$$\text{simple interest} = \frac{86\times15\times4}{100} = 51.60$$

Amount = Principal + simple interest

= 86 + 51.60 = ₹137.60

34. The amount of ₹800 becomes ₹956 in 3 years at a fixed rate of simple interest. If the rate of interest increases by 4% then the amount of ₹800 will be increased in 3 years?

(a) ₹1025 (b) ₹1020.80
(c) ₹1052 (d) ₹1054

RRB RPF SI – 10/01/2019 (Shift-I)

Ans : (c) simple interest = Amount – Principal

$$\text{Simple interest} = \frac{P\times R\times T}{100}$$

$$156 = \frac{800\times\text{Rate}\times3}{100}, \quad \text{Rate} = \frac{156}{24} = 6.5\%$$

When rate increase 4% then

Amount = simple interest + Principal

$$= \frac{800\times10.5\times3}{100} + 800 = 8\times10.5\times3 + 800$$

= 84 × 3 + 800 = 1052

35. Ganesh took a loan of ₹14000 which is to be repaid after three years at compound interest at the rate of 10% per annum. What is the total amount he will have to pay after three years?

(a) ₹16200 (b) ₹18634
(c) ₹17940 (d) ₹18497

RRB JE - 31/05/2019 (Shift-III)

Ans. (b) Principal = ₹14000, time = 3 Year

Rate = 10% Annual

$$\text{Amount} = 14000\left(1+\frac{10}{100}\right)^3$$

$$= 14000\times\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10} = ₹18634$$

36. Calculate 1.5 years of interest at a 10% annual interest rate on an amount of ₹1000 when interest is calculated on a half yearly compound interest basis.

(a) ₹167.36 (b) ₹157.63
(c) ₹150.25 (d) ₹160.55

RRB JE - 22/05/2019 (Shift-I)

Ans : (b) P = ₹1000, r = 10%, n = 1.5 Year = 3 Half year

$$CI = P\left(1+\frac{r}{200}\right)^n - P$$

$$= 1000\left(1+\frac{10}{200}\right)^3 - 1000$$

$$= 1000\times\frac{21}{20}\times\frac{21}{20}\times\frac{21}{20} - 1000$$

$$= \frac{9261}{8} - 1000 = 1157.63 - 1000 = ₹\ 157.63$$

37. When calculated by compound interest on a half yearly basis, find the compound interest of 18 months at 40% per annum on an amount of ₹18500.

(a) ₹ 13468 (b) ₹ 16280
(c) ₹ 16000 (d) ₹ 15469

RRB RPF Constable – 25/01/2019 (Shift-I)

Ans : (a) P = ₹ 18500
R (Annual) = 40%
Half yearly Rate = r/2% = 20%
n = 3 [18 month = 3 Half year]

$$CI = P\left(1+\frac{r}{100}\right)^n - P$$

$$= 18500\left(1+\frac{120}{100}\right)^3 - 18500$$

$$= 18500\left(\frac{6}{5}\right)^3 - 18500$$

$$= 18500\times\frac{216}{125} - 18500$$

= 31968 – 18500 = ₹13468

38. At a rate of 10% half yearly compound interest at ₹6000. Find the compound interest received in $1\frac{1}{2}$ years

(a) ₹870 (b) ₹900
(c) ₹946 (d) ₹910

RRB JE - 27/06/2019 (Shift-III)

Ans : (c) ∵ Interest is half yearly

$$\therefore\ R = \frac{10}{2} = 5\%$$

$$n = 1\frac{1}{2} = \frac{3}{2}\times 2 = 3 \text{ half yearly}$$

$$C.I. = P\left[\left(1+\frac{R}{100}\right)^n - 1\right]$$

$$C.I. = 6000\left[\left(1+\frac{5}{100}\right)^3 - 1\right]$$

$$C.I. = 6000\times\frac{(9261-8000)}{8000}$$

$$C.I. = 6\times\frac{1261}{8} = \frac{7566}{8} = 945.75$$

Compound interest (approximate) = ₹946

39. Divide the amount of ₹3364 between A and B in such a way that the amount received after 5 years of investing A's share at the rate of 5% compound interest annually, after 7 years of investing B's share at the same rate will be equal to the amount received. Find the share of A.

(a) ₹1564 (b) ₹1600
(c) ₹1764 (d) ₹1864

RRB JE - 27/06/2019 (Shift-I)

Ans : (c) Let A's share = ₹x
∴ B's share = ₹(3364 – x)
Amount of A -

$$\text{Amount} = \text{Principal}\left(1+\frac{r}{100}\right)^t$$

$$= x\left(1+\frac{5}{100}\right)^5$$

$$= x\left(\frac{21}{20}\right)^5$$

Amount of B –

$$\text{Amount} = (3364 - x)\left(1+\frac{5}{100}\right)^7$$

$$= (3364 - x)\left(\frac{21}{20}\right)^7$$

$$x\left(\frac{21}{20}\right)^5 = (3364 - x)\left(\frac{21}{20}\right)^7$$

$$x = (3364 - x)\ \frac{21}{20}\times\frac{21}{20}$$

400x = 3364 × 441 – 441x
841x = 3364 × 441

$$x = \frac{3364\times 441}{841} = ₹1764$$

40. In what time will the amount of ₹1000 become ₹1331 at the rate of 10% annual compound interest?

(a) 4 Year (b) 3 Year
(c) 2 Year (d) 5 Year

RRB JE - 27/06/2019 (Shift-I)

Ans : (b) Principal (P) = ₹1000
Amount (A) = ₹1331
Rate (r) = 10%
Time (t) = ?

$$A = P\left(1+\frac{r}{100}\right)^t, \quad 1331 = 1000\left(1+\frac{10}{100}\right)^t$$

$$\frac{1331}{1000} = \left(\frac{11}{10}\right)^t$$

$$\left(\frac{11}{10}\right)^3 = \left(\frac{11}{10}\right)^t$$

On comparing of powers,
t = 3 Year

41. Meena took a car loan of ₹275000 from the bank. She paid interest at an annual rate of 8% (p.a.) and paid full amount in 3 years While paying, she paid her old scooter and ₹335000 to the bank. What was the total price of the scooter?

(a) ₹60, 000 (b) ₹6,000
(c) ₹66,000 (d) ₹6,600

RRB RPF Constable – 20/01/2019 (Shift-II)

Ans : (b)
Loan amount taken by meena (P) = ₹275000
Rate(r) =8%, time n = 3Year
Cost of scooter = Amount-335000

$$= 275000 + \frac{275000\times 8\times 3}{100} - 335000$$

$$= 275000 + 66000 - 335000$$

$$= 341000 - 335000 = ₹6000$$

Hence, cost of scoter is ₹6000

42. What will be the amount of ₹3000 after 2 years if compound interest is compounded annually at about 12 percent per annum interest rate?

(a) ₹3,763 (b) ₹3,773
(c) ₹3,873 (d) ₹3,766

RRB RPF Constable – 19/01/2019 (Shift-I)

Ans. (a) : Principal = ₹3000, Rate = 12%, Time = 2Year

$$\therefore \quad \text{Amount} = \text{Principal}\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}$$

$$= 3000\left(1+\frac{12}{100}\right)^2$$

$$= 3000\times\frac{28}{25}\times\frac{28}{25}$$

$$= \frac{784\times 24}{5}$$

$$= 3763.2 \approx ₹\ 3763$$

43. Satya invested some amount in a fixed deposit what amount of money would be receive on maturity if he invested ₹14500 at a 20% compound interest rate at quarterly compounding rate for a period of 6 months.

(a) ₹ 15,986.25 (b) ₹ 15, 986.5
(c) ₹ 15,986.35 (d) ₹ 15,986

RRB NTPC 17.01.2017 Shift-3

Ans : (a) Invested amount = ₹14500
Time = 6 month = 2 quarter

Rate = 20% Annual $= \frac{20}{4}$% quarter = 5% quarter

$$\because\ A = P\left(1+\frac{r}{100}\right)^n$$

Received amount on maturity to satya

$$= 14500\left(1+\frac{5}{100}\right)^2 = 14500\left(\frac{21}{20}\right)^2 = 14500\times\frac{441}{400}$$

= ₹15986.25

44. Neha's ₹8000 amount becomes ₹9200 after 3 years at a fixed simple interest rate. If the interest rate was increased by 2%, what amount she would have received?

(a) ₹8900 (b) ₹9800
(c) ₹2000 (d) ₹9680

RRB Group-D – 22/10/2018 (Shift-III)

Ans : (d) Amount = Principal + Simple interest

$$9200 = 8000 + \frac{8000\times 3\times r}{100}$$

1200 = 80 × 3r
r = 5%
New Rate = 5 + 2 = 7%

$$\text{simple interest} = \frac{8000\times 3\times 7}{100}$$

= 1680
Total amount = 8000 + 1680 = ₹9680

45. An amount of 200 was invested in a scheme for one year, offering a simple interest of 10% per annum, compounding the interest annually. Another amount of ₹200 was invested for one year in a scheme at 10% per annum, but interest was compounded half yearly. How much will be the interest earned under the second plan?

(a) 50 paisa (b) ₹1
(c) 10 paisa (d) 25 paisa

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (a) Principal = ₹ 200, Rate = 10%, time = 1 Year

$$\text{Compound amount of first plan} = 200\left(1+\frac{10\times 1}{100}\right)$$

$$= \frac{200\times 11}{10} = 220$$

Compound amount of second plan

$$= \text{Principal}\left(1+\frac{\text{Rate}}{100}\right)^{\text{time}}$$

Principal = ₹ 200

Rate $= \frac{10\%}{2} = 5\%$ (Half yearly Compound interest)

time = 2 Half yearly

Compound amount $= 200 \times \left(1 + \frac{5}{100}\right)^2$

$= \frac{200 \times 21 \times 21}{20 \times 20} = \frac{441}{2} =$ ₹220.5

Profit earned under second plan = amount of second plan – amount of first plan = 220.50 – 220 = 50 paisa

46. Manoj invested ₹15000 in a fixed deposit scheme for 3 years, combining 5% per annum annually. What amount will Manoj get on maturity of fixed deposit.

(a) ₹13,764, .37 (b) ₹17,463.37
(c) ₹17,643.37 (d) ₹17,364.37

RRB Group-D – 26/10/2018 (Shift-II)

Ans : (d)

Compound amount $= \text{Principal}\left(1 + \frac{\text{Rate}}{100}\right)^{\text{time}}$

$= 15000\left(1 + \frac{5}{100}\right)^3$

$= 15000 \times \frac{21}{20} \times \frac{21}{20} \times \frac{21}{20}$

$= \frac{138915}{8} =$ ₹ 17364.37

So on maturity of fixed deposit manoj will get ₹ 17364.37

47. Due to an intense campaign against smoking, the percentage of smokers in an area is falling by 10% every year as compared to previous year. If currently the numbers of smokers is 8748, then what was the number of smokers 3 years ago?

(a) 12000 (b) 16253
(c) 11643 (d) 10000

RRB Group-D – 24/10/2018 (Shift-II)

Ans. (a) : From the Formula–

For the population of n year ago

$A = \frac{P}{\left(1 - \frac{r}{100}\right)^n}$

$\because$ Given–

P = 8748 r = 10%

n = 3 Year

$\therefore A = \frac{8748}{\left(1 - \frac{10}{100}\right)^3} = \frac{8748}{\left(\frac{9}{10}\right)^3}$

$A = 8748 \times \frac{10}{9} \times \frac{10}{9} \times \frac{10}{9} = 12000$

48. Sita borrowed ₹180000 at a simple interest rate of 10% per annum. On the same day, she gave that amount to her friend at an annual compound interest rate. How much rupee did she gain at the end of 2 years?

(a) ₹ 2,000 (b) ₹ 1,600
(c) ₹ 2,200 (d) ₹ 1,800

RRB Group-D – 15/10/2018 (Shift-III)

Ans. (d) : Rate = 10%, Principal = ₹ 180000,

Time = 2 Year

Simple interest $= \frac{P \times R \times T}{100}$

$= \frac{180000 \times 10 \times 2}{100}$

= ₹ 36000

Compound amount $= \text{Principal}\left(1 + \frac{\text{Rate}}{100}\right)^{\text{time}}$

Compound Amount $= 180000 \times \left(1 + \frac{10}{100}\right)^2$

$= 180000 \times \frac{11}{10} \times \frac{11}{10}$

$= 1800 \times 121$

= ₹ 217800

Compound interest = Compound Amount - Principal

= 217800 – 180000

= ₹ 37800

Gain profit in two year

= Compound interest – simple interest

= 37800 – 36000 = ₹ 1800

49. Mani deposits ₹8000 in a bank on which he gets 5% annual interest. If the interest is calculated annually, then after two years, what will be the amount in his account?

(a) ₹ 8500 (b) ₹ 8700
(c) ₹ 8820 (d) ₹ 8600

RRB Group-D – 09/10/2018 (Shift-II)

Ans. (c) :

Compound amount $= \text{Principal}\left(1 + \frac{\text{Rate}}{100}\right)^{\text{time}}$

$= 8000\left(1 + \frac{5}{100}\right)^2$

$= 8000 \times \frac{21}{20} \times \frac{21}{20}$

$= 20 \times 441 =$ ₹8820

50. If the interest is calculated annually, then the amount of ₹2000 will be approximately after 3 years at the rate of 10% compound annual interest?

(a) ₹ 2510 (b) ₹ 2662
(c) ₹ 2520 (d) ₹ 2726

RRB Group-D – 09/10/2018 (Shift-II)

Ans. (b) :

Compound amount $= \text{Principal}\left(1+\frac{\text{Rate}}{100}\right)^{\text{time}}$

$= 2000\times\left(1+\frac{10}{100}\right)^3$

$= 2000\times\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10}$

$= 2\times1331$

= ₹ 2662

51. A person takes ₹32000 at a simple interest rate of 9% per annum and deposits it in the bank at a compound interest rate of 10% per annum. What will be the total increase in his wealth by the end of the third year?

(a) ₹1,940 (b) ₹1,952

(c) ₹926 (d) ₹2,904

RRB Group-D – 01/10/2018 (Shift-III)

Ans : (b) simple interest $= \frac{32000\times9\times3}{100} = 27\times320$

simple interest (S.I.) = ₹ 8,640

Compound amount $= 32000\left(1+\frac{10}{100}\right)^3$

$= 32000\left(\frac{11}{10}\right)^3$

$= 32000\times\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10}$

$= 32\times1331$

= ₹ 42,592

Compound interest = Compound amount – Principal

= 42592 – 32000

= ₹ 10,592

Growth in amount

= Compound interest – simple interest

= 10,592 – 8,640

= ₹ 1,952

52. Z paid ₹10920 after 4 years with 10% annual simple interest. How much money did he borrow?

(a) ₹7,600 (b) ₹7,800

(c) ₹8,200 (d) ₹7,400

RRB NTPC 05.04.2016 Shift-1

Ans : (b) Given-

Rate = 10% Annual, time = 4 Year

Amount = ₹10920

Suppose ₹x was borrowed by Z

As per the question-

$x+\frac{x\times10\times4}{100}=10920$

$\Rightarrow \frac{7x}{5}=10920$

$\Rightarrow 7x = 10920\times5$

$\Rightarrow x = \frac{10920\times5}{7} = 1560\times5$ = ₹7800

53. A man deposits ₹ 500 at the beginning of each year for 2 years at 10% annually compound interest rate. Find the maturity amount at the end of the 2nd year.

(a) ₹1.050 (b) ₹1,150

(c) ₹1,155 (d) ₹1,200

RRB NTPC 05.04.2016 Shift : 3

Ans : (c)

Amount at end of first year $= 500\times\left(1+\frac{10}{100}\right)^1$

$= 500\times\frac{11}{10} = 550$

Principal for second year $= 500+550 = 1050$

Amount at end of second year $= 1050\times\left(1+\frac{10}{100}\right)^1$

$= 1050\times\frac{11}{10} = 1155$

So maturity amount at end of 2year = ₹1155

54. A bank pays interest on a saving account at half the rate of 4%. What will be the effective interest rate at the end of the year?

(a) 4.04% (b) 4.01%

(c) 4.02% (d) 4.00%

RRB NTPC 31.03.2016 Shift : 1

Ans : (a) The rate of interest is combined for half yearly.

∴ for one Year,

Time = 2 Half year

Rate $= \frac{4}{2} = 2\%$ Half yearly

∴Amount $(A) = 100\left(1+\frac{2}{100}\right)^2$ [Let Principal =100]

$= 100\times\frac{51}{50}\times\frac{51}{50}$ = 104.04 रु.

∴ Effective interest rate at the end of year

= 104.04–100 = 4.04%

55. How much will the amount become while ₹37500 are invested at an annual compound interest rate of 8% in $1\frac{1}{2}$ years when interest rates are compounded half-yearly?

(a) ₹42,182.40 (b) ₹42,000
(c) ₹42,120 (d) ₹42,812.40

RRB NTPC 28.03.2016 Shift : 1

Ans : (a) Given-
Principal (P) = ₹ 37500, r = 8%
Annual = 4% Half yearly

$$\text{Time (t)} = 1\frac{1}{2}\text{ Year} = 3\text{ Half year}$$

$$\text{Amount}(A) = P\left(1+\frac{r}{100}\right)^t$$

$$= 37500\left(1+\frac{4}{100}\right)^3$$

$$= 37500\times\frac{26}{25}\times\frac{26}{25}\times\frac{26}{25} = ₹42182.40$$

56. 'A' invest ₹10000 in a scheme at a special rate. At the end of two years the amount becomes ₹11664. Find the annual interest rate if compound interest rate is applicable.

(a) 7.9 (b) 8.5
(c) 8.7 (d) 8.0

RRB NTPC 18.01.2017 Shift : 2

Ans : (d) Compound amount (A) = Principal

$$(P)\times\left(1+\frac{\text{Rate}}{100}\right)^{\text{time}}$$

$$\Rightarrow 11664 = 10000\left(1+\frac{R}{100}\right)^2$$

$$\Rightarrow \frac{11664}{10000} = \left(1+\frac{R}{100}\right)^2$$

$$\Rightarrow \frac{729}{625} = \left(1+\frac{R}{100}\right)^2$$

$$\Rightarrow 1+\frac{R}{100} = \frac{27}{25}$$

$$\Rightarrow \frac{R}{100} = \frac{2}{25}$$

$$\boxed{R = 8\%}$$

57. If Ram deposits ₹2000 in his saving account on which he gets 20% annual interest at the compound annual interest rate. How much will be amount in his account after one year (the interest calculated in half yearly)?

(a) ₹3530 (b) ₹ 2420
(c) ₹2630 (d) ₹ 3870

RRB NTPC 18.01.2017 Shift : 3

Ans : (b) one Year = 2 Half year

$$\text{Rate} = \frac{20}{2} \Rightarrow 10\%$$

$$\text{Compound amount} = 2000\times\left(1+\frac{10}{100}\right)^2$$

$$= 2000\times\frac{11}{10}\times\frac{11}{10} = ₹2420$$

58. ₹10000 was given at the rate of 20% compound interest per annum. If this interest is calculated half early, find the compound money after 2 years

(a) ₹14600 (b) ₹12500
(c) ₹14642 (d) ₹14641

RRB NTPC 16.04.2016 Shift : 1

Ans : (d) Interest calculated half yearly

$$\text{Rate} = \frac{20}{2} = 10\%$$

and time = 2 × 2 = 4 Year

$$\therefore \text{Amount} = \text{Principal}\left(1+\frac{\text{Rate}}{100}\right)^{\text{time}}$$

$$= 10000\left(1+\frac{10}{100}\right)^4$$

$$= 10000\left(1+\frac{1}{10}\right)^4 = 10000\left(\frac{11}{10}\right)^4$$

$$= 10000\times\frac{14641}{10000} = ₹14641$$

59. In what time will the principal of ₹10000 be increased to ₹13310 at the interest rate of 10% per annum in a bank when interest is compounded annually?

(a) 4 Year (b) 5 Year
(c) 3 Year (d) 6 Year

RRB NTPC 16.04.2016 Shift : 2

Ans : (c) Given-
Principal (P) = ₹10000
Amount (A) = ₹13310
Rate (R) = 10% Annual
time (t) = ?

Formula- $A = P\left(1+\frac{R}{100}\right)^t$ from-

$$13310 = 10000\left(1+\frac{10}{100}\right)^t$$

$$\left(\frac{11}{10}\right)^3 = \left(1+\frac{10}{100}\right)^t$$

$$\left(\frac{11}{10}\right)^3 = \left(\frac{11}{10}\right)^t$$

On comparing the powers of both sides -
t = 3 Year

60. Mr. Vagish invested money in Ekdi. How much amount will he get at maturity if ₹4500 is invested for 6 months at a compound interest rate of 20% annually and the interest is compounded quarterly?

(a) ₹4961.5 (b) ₹4961.25
(c) ₹4961.35 (d) ₹4961

RRB NTPC 11.04.2016 Shift : 3

Ans : (b) Rate (r) $= \frac{20}{4} = 5\%$

time (n) = 2 quarter

Amount (A) = $P\left(1+\frac{r}{100}\right)^n = 4500\left(1+\frac{5}{100}\right)^2$

$= 4500 \times \frac{21}{20} \times \frac{21}{20} = 4961.25$ ₹

61. Mr. Nishi barrows ₹2000 at an annual compound interest rate of 5% what will be the compound interest for 2 years when the interest is compounded annually.

(a) ₹205 (b) ₹2205
(c) ₹2250 (d) ₹250

RRB NTPC 07.04.2016 Shift : 1

Ans : (a) Rate = 5%, Time = 2 Year,
Principal = 2000

Amount = Principal $\left(1+\frac{\text{Rate}}{100}\right)^{\text{time}} = 2000\left(1+\frac{5}{100}\right)^2$

$= 2000 \times \left(\frac{21}{20}\right)^2 \Rightarrow 2000\left(\frac{21 \times 21}{20 \times 20}\right) = 2205$

Interest = Amount − Principal = 2205 – 2000 = 205

62. Prachi barrows ₹550 at a simple annual interest rate of 5%. How much will she have to pay after 4 years?

(a) 101 (b) 606
(c) 660 (d) 110

RRB NTPC 07.04.2016 Shift : 3

Ans :(c) Principal (P) = 550
Rate (R) = 5%
time (T) = 4 Year

simple interest (SI) $= \frac{PRT}{100} = \frac{550 \times 5 \times 4}{100} = 110$

Amount = Principal + interest = 550 + 110 = 660

63. A person named Shri Ram invested ₹14000 in FD (fixed deposit) what amount will he get after the maturity period if he invests the amount (capital) for 6 months at quarterly basis at a compound interest rate of 20% per annum?

(a) ₹ 15,437 (b) ₹ 15,434
(c) ₹ 15,436 (d) ₹ 15,435

RRB NTPC 19.01.2017 Shift : 3

Ans : (d) Principal (P) = ₹14000
When rate pay is quarter then rate

$= \frac{\text{Annual rate}}{4} = \frac{20}{4} = 5\%$

time (n) = 6 month = 2 quarter

$A = P\left(1+\frac{R}{100}\right)^n$

$A = 14000\left(1+\frac{5}{100}\right)^2$

$A = 14000 \times \left(\frac{21}{20}\right)^2$

$\Rightarrow 14000 \times \frac{441}{400} = 441 \times 35 = \boxed{15435}$

64. At simple interest for the money ₹1500, the compound money becomes ₹1800 in two years, If the rate of interest is increased by 5%, what will be the compound money.

(a) ₹1500 (b) ₹1900
(c) ₹ 1950 (d) ₹2000

RRB NTPC 22.04.2016 Shift : 1

Ans : (c) Given −
Principal = ₹1500
time = 2 Year
Amount = ₹1800
∴ Interest = Amount – Principal
= 1800 -1500 = ₹300

Formula- Simple interest $= \frac{P \times R \times T}{100}$

$\Rightarrow 300 = \frac{1500 \times R \times 2}{100}$

$\Rightarrow R = 10\%$

after increasing interest rate by 5% = 10% + 5% = 15%

New Simple interest $= \frac{1500 \times 15 \times 2}{100} = 450$

New Amount = Principal + interest
= 1500 + 450
= $\boxed{₹1950}$

65. Mr. Yashwant invested a few rupees in FD. What will be the total amount on maturity if ₹10000 is invested for 6 months at a rate of 20% compound interest annually, compounded quarterly?

(a) 11025.25 (b) 11025
(c) 11025.75 (d) 11025.5

RRB NTPC 22.04.2016 Shift : 2

Ans : (b) Principal (P) = 10000, r = 20% Annual
$= \frac{20}{4}$% quarter, r = 5% quarter

(n) Time = 6 month,
= 2 quarter

Amount $= P\left(1+\frac{r}{100}\right)^n$

$= 10,000\left(1+\frac{5}{100}\right)^2$

$= 10,000\left(1+\frac{1}{20}\right)^2$

$= 10,000\left(\frac{21}{20}\right)^2$

$= 10,000 \times \frac{21}{20} \times \frac{21}{20} = ₹11025$

66. Sanjeev invests in a fixed deposit. If ₹11000 is invested at the rate of 20% compound interest annually for 6 months and the interest is compounded quarterly, what is the total amount he will get on maturity?

(a) ₹12127.25 (b) ₹12127.50
(c) ₹12127.75 (d) ₹12127

RRB NTPC 26.04.2016 Shift : 2

Ans : (b) Interest is compounded in quarterly,

$$\text{Time} = \frac{6}{12} \times 4 = 2 \text{ quarter}$$

$$\text{Rate} = \frac{20}{4} = 5\%$$

$$\text{Amount } (A) = P\left(1+\frac{r}{100}\right)^n$$

$$= 11000\left(1+\frac{5}{100}\right)^2$$

$$= 11000 \times \frac{21}{20} \times \frac{21}{20}$$

$$= 12127.5$$

Type - 3

67. Ranjan borrows ₹7500 at an annual compound interest rate of 4%. What will be the compound interest for 2 years while the interest is compounded annually?

(a) ₹612 (b) ₹8112
(c) ₹8121 (d) ₹621

RRB RPF SI – 11/01/2019 (Shift-II)

Ans : (a) P = 7500, R = 4%, n = 2 Year

$$\text{C.I.} = P\left[\left(1+\frac{r}{100}\right)^n - 1\right]$$

$$= 7500\left[\left(1+\frac{4}{100}\right)^2 - 1\right] = 7500\left[\left(\frac{26}{25}\right)^2 - 1\right]$$

$$= 7500 \times \frac{676-625}{625} = 7500 \times \frac{51}{625} = 612 ₹$$

68. Find the compound interest on ₹ 15,625 for 1 year 6 months at 8% compound interest rate per annum, when compounded half-yearly?

(a) ₹1,951 (b) ₹1,950
(c) ₹1,900 (d) ₹1,952

RRB Group-D – 12/10/2018 (Shift-III)

Ans : (a) Principal = ₹ 15,625

Rate = 8% Annual ⇒ 4% Half yearly

Time = 1 Year 6 months = 3 Half year

$$\text{Compound interest} = 15,625\left(1+\frac{4}{100}\right)^3 - 15625$$

$$= 15625 \times \frac{26}{25} \times \frac{26}{25} \times \frac{26}{25} - 15625$$

$$= 17,576 - 15625 = 1951$$

69. What will be the compound interest for $2\frac{3}{4}$ years at 8% compound interest rate per annum on ₹31250?

(a) ₹ 7300 (b) ₹ 7800
(c) ₹ 7337 (d) ₹ 7387

RRB Group-D – 22/09/2018 (Shift-I)

Ans : (d) $A = P\left(1+\frac{R}{100}\right)^2\left(1+\frac{3R}{400}\right)$

$$A = 31250 \times \left(1+\frac{8}{100}\right)^2\left(1+\frac{3\times 8}{400}\right)$$

$$A = 31250 \times \frac{27}{25} \times \frac{27}{25} \times \frac{106}{100}$$

$$A = 27 \times 27 \times 53$$

$$A = ₹\ 38637$$

∴ C.I. (Compound interest) = 38637 – 31250 = ₹ 7387

70. On investing a sum of money at a 4% annual rate of compound interest yields ₹78030 at the end of 1 year, when the interest is compounded half yearly. The fund is–

(a) ₹76,000 (b) ₹71,400
(c) ₹72,500 (d) ₹75,000

RRB Group-D – 04/12/2018 (Shift-III)

Ans. (d) when interest compounded half yearly then,

$$r = \frac{4}{2}\% = 2\%$$

time = 1 Year = 2 Half yearly

$$\text{Amount} = P\left(1+\frac{r}{100}\right)^t$$

$$78030 = P\left(1+\frac{2}{100}\right)^2$$

$$78030 = P\left(1+\frac{1}{50}\right)^2$$

$$\left(\frac{51}{50}\right)^2 P = 78030$$

$$P = \frac{78030 \times 2500}{2601}$$

$$P = ₹75000$$

71. The interest received on a fixed amount at a rate of 10% in a year is ₹400. Compound interest for the same amount at the same rate and for the same period if the interest is compounded half yearly will be–

(a) ₹ 400 (b) ₹ 210
(c) ₹ 410 (d) ₹ 200

RRB Group-D – 05/11/2018 (Shift-I)

Ans. (c) Let Amount = ₹ P, Rate = 10%, T = 1 Year

∴ As per the question,

$$400 = \frac{p \times r \times t}{100}$$

$$400 = \frac{p \times 10 \times 1}{100}$$

P = ₹ 4000

R = 10%, t = 2 Half yearly

Compound amount = $P\left(1+\frac{r}{2\times100}\right)^2$

$$\Rightarrow P\left(1+\frac{r}{2\times100}\right)^2$$

$$\Rightarrow 4000\left(1+\frac{10}{200}\right)^2$$

$$\Rightarrow 4000\left(\frac{21}{20}\right)^2$$

$$\Rightarrow 4000\times\frac{21}{20}\times\frac{21}{20}$$

⇒ ₹ 4410

Compound interest = Amount − Principal

Compound interest = 4410 – 4000 = ₹410

72. The simple interest on a sum of money for 3 years at 12% simple interest rate per annum is ₹6750. What will be the compound interest on the same amount at the same rate for the same period when compounded annually?

(a) ₹ 7,092.40 (b) ₹ 7,000

(c) ₹ 7,592.40 (d) ₹ 7,500.40

RRB Group-D – 12/10/2018 (Shift-I)

Ans. (c) : $SI = \frac{PRT}{100}$

From,

$$6750 = \frac{P\times12\times3}{100}$$

$$P = \frac{6750\times100}{12\times3}$$

$$P = 18750$$

Compound amount $(A) = P\left(1+\frac{R}{100}\right)^n$

$$A = 18750\left(1+\frac{12}{100}\right)^3$$

$$A = 18750\times\frac{28}{25}\times\frac{28}{25}\times\frac{28}{25}$$

A = ₹ 26342.4

Compound interest = Amount (A) - Principal (P)

= 26342.4-18750 = ₹7592.4

73. Any money is deposited at 8% per annum compound interest rate. If the interest for the first year is 72, then get the interest for the second year.

(a) ₹77.56 (b) ₹ 77.64

(c) ₹ 77.76 (d) ₹ 85.77

RRB NTPC 17.01.2017 Shift-2

Ans : (c) $P = \frac{S.I.\times100}{R\times T}$

Amount $= \frac{72\times100}{8\times1} = 900$

amount for second year = 900 + 72 = 972

Interest of second year $= \frac{972\times8\times1}{100} = 77.76$

74. What will be the compound interest of ₹172000 3 years at the of 8% compound interest rate per annum.

(a) ₹44,670 (b) ₹11,667

(c) ₹41,280 (d) ₹46,470

RRB NTPC 04.04.2016 Shift : 2

Ans : (a) Given- Principal (P) = 172000,

Rate = 8% Annual

Time (n)=3 Year

$$C.I = P\left(1+\frac{r}{100}\right)^n - P = P\left[\left(1+\frac{r}{100}\right)^n - 1\right]$$

$$= 172000\left[\left(1+\frac{8}{100}\right)^3 - 1\right]$$

$$= 172000\left[\left(\frac{27}{25}\right)^3 - 1\right]$$

$$= 172000\left[\frac{19683}{15625} - 1\right]$$

$$= 172000\times\frac{19683-15625}{15625}$$

$$= 11.008\times4058 = 44,670$$

75. Find the compound interest at ₹7500 at the rate of 12% per annum for 2 years 4 months to the nearest rupee, this interest is calculated on an annual basis?

(a) 2284 (b) 2176

(c) 2097 (d) 2235

RRB NTPC 03.04.2016 Shift : 1

Ans : (a) Principal P = ₹ 7500 , $t = 2\frac{1}{3}$ Year, r = 12% Annual

Compound amount $A = P\left(1+\frac{r}{100}\right)^t$

$$= 7500\left(1+\frac{12}{100}\right)^{2\frac{1}{3}}$$

$$= 7500\left(1+\frac{3}{25}\right)^2\left(1+\frac{3}{25}\right)^{\frac{1}{3}}$$

$$= 7500\times\frac{28}{25}\times\frac{28}{25}\times\left(1+\frac{1}{3}\times\frac{3}{25}\right)$$

$$= 7500\times\frac{28}{25}\times\frac{28}{25}\times\frac{26}{25} = ₹\ 9784.32$$

Compound interest = ₹ 9784.32 – 7500 = ₹ 2284.32

So Compound interest nearest will be ₹2284

76. Find the interest on ₹ 25000 at 12% per annum compounded annually for 3 years

(a) ₹9,000 (b) ₹9,833.40
(c) ₹10123.20 (d) ₹10,678.90

RRB NTPC 03.04.2016 Shift : 2

Ans : (c) Compound interest $= 25000\left[\left(1+\frac{12}{100}\right)^3 - 1\right]$

$= 25000\left[\left(1+\frac{3}{25}\right)^3 - 1\right]$

$= 25000\left[\left(\frac{28}{25}\right)^3 - 1\right]$

$= 25000\left(\frac{21952}{15625} - 1\right)$

$= 25000 \times \frac{6327}{15625} = ₹10123.20$

77. What will be the compound interest at ₹24000 in 2 years at 25% per annum if the interest is compounded annually.

(a) ₹37,500 (b) ₹13,500
(c) ₹38,400 (d) ₹36,400

RRB NTPC 03.04.2016 Shift : 3

Ans : (b) Compound interest

$(C.I.) = P\left(1+\frac{r}{100}\right)^n - P$

$= 24000\left(1+\frac{25}{100}\right)^2 - 24000$

$= 24000\left(1+\frac{1}{4}\right)^2 - 24000$

$= 24000\left(\frac{5}{4}\right)^2 - 24000 \quad = 24000\left[\frac{25}{16} - 1\right]$

$= 24000\left(\frac{25-16}{16}\right) = 24000 \times \frac{9}{16} = 13,500$

78. If the interest is compounded annually then for 2 years at ₹48000, what will be the compound interest at 20% per annum?

(a) ₹69,120 (b) ₹21,120
(c) ₹76,800 (d) ₹72,000

RRB NTPC 02.04.2016 Shift : 2

Ans : (b) P(Principal) = 48000, Rate (r) = 20%
time (T) = 2 years

Amount $= P\left(1+\frac{r}{100}\right)^T = 48000\left(1+\frac{20}{100}\right)^2$

$= 48000\left(1+\frac{1}{5}\right)^2 = 48000\left(\frac{6}{5}\right)^2$

$= 48000 \times \frac{6}{5} \times \frac{6}{5} = 69120$

interest = Amount – Principal
= 69120 – 48000 = 21120

79. How much will become ₹40,000 at 2% per annum at an annual rate of 20%, if interest is compounded annually?

(a) ₹48,620 (b) ₹58,564
(c) ₹57,600 (d) ₹60,000

RRB NTPC 02.04.2016 Shift : 3

Ans : (c) Given- Principal (P) = 40,000
time (n) = 2 Year
Rate=20% Annual

$A = P\left(1+\frac{r}{100}\right)^n$

$= 40000\left(1+\frac{20}{100}\right)^2$

$= 40000 \times \frac{6}{5} \times \frac{6}{5} = 57600$

80. If Jahnavi borrows ₹1,25,000 at an interest of 8% per annum, what will be the amount that is payable at the end of 2 years?

(a) ₹1,45,800 (b) ₹2,00,000
(c) ₹1,45,000 (d) ₹1,35,800

RRB NTPC 28.03.2016 Shift : 2

Ans :(a) P = 125000, R = 8%, n = 2 Year, A = ?

$A = P\left(1+\frac{r}{100}\right)^n$

$A = 125000\left(1+\frac{8}{100}\right)^2$

$= 125000 \times \frac{108 \times 108}{100 \times 100}$

= 25 ×54 × 108 = ₹145,800

81. 20% compound interest is being given every year at ₹10000. Calculate the amount received after 2 years if interest is calculated charged half yearly.

(a) ₹10041 (b) ₹14641
(c) ₹12000 (d) ₹13660

RRB NTPC 19.04.2016 Shift : 2

Ans : (b) Given–
P = ₹10000

$r = \frac{20}{2} = 10\%$ Half yearly

∵ 1 Year = 2 Half year
∴ 2 Year = 4 Half year

$A = P\left(1+\frac{r}{100}\right)^t$ from-

$A = 10000\left(1+\frac{10}{100}\right)^4 = 10000\left(\frac{11}{10}\right)^4$

$= \frac{1331 \times 11 \times 10000}{10000} = ₹14641$

82. Find the compound interest of ₹5000 for 3 years at an annual compound interest rate of 10%.

(a) 1655 (b) 1500
(c) 1600 (d) 1800

RRB NTPC 12.04.2016 Shift : 1

Ans : (a) Principal (P) = ₹5000
time (t) = 3 Year
Annual interest Rate (R) = 10%

$$\text{Compound interest } (C.I.) = P\left[\left(1+\frac{R}{100}\right)^t - 1\right]$$

$$= 5000\left[\left(1+\frac{10}{100}\right)^3 - 1\right]$$

$$= 5000\left[\left(\frac{11}{10}\right)^3 - 1\right]$$

$$= 5000\left[\frac{1331-1000}{1000}\right]$$

$$= 5000 \times \frac{331}{1000} = 1655$$

83. Ibrahim barrowed ₹7500 at an annual compound interest rate of 5%. What will be the compound interest after 2 years which is compounded annually.

(a) ₹768.75 (b) ₹8268.75
(c) ₹8286.75 (d) ₹786.75

RRB NTPC 11.04.2016 Shift : 1

Ans : (a) Principal (P) = ₹7500
Annual Rate (R) = 5%
time (t) = 2 Year

$$\text{Compound interest } = P\left[\left(1+\frac{R}{100}\right)^t - 1\right]$$

$$= 7500\left[\left(1+\frac{5}{100}\right)^2 - 1\right]$$

$$= 7500\left[\left(\frac{21}{20}\right)^2 - 1\right] = 7500\left[\frac{441}{400} - 1\right]$$

$$= 7500\left[\frac{441-400}{400}\right] = 7500 \times \frac{41}{400}$$

= ₹768.75

84. Devesh borrowed ₹4,500 at an annual rate of 4% compound interest. What will be the compound interest after 2 years if the interest is compounded annually.

(a) ₹367.2 (b) ₹4,867.2
(c) ₹4,876.2 (d) ₹376.2

RRB NTPC 11.04.2016 Shift : 2

Ans : (a) P = 4500, R = 4%, n = 2 Year

$$A = P\left(1+\frac{R}{100}\right)^n$$

$$= 4500\left(1+\frac{4}{100}\right)^2$$

$$= 4500\left(\frac{104}{100}\right)^2$$

$$= 4500 \times \frac{104 \times 104}{100 \times 100} = 4867.2$$

CI = A – P
= 4867.2 – 4500 = ₹367.2

85. Harsh borrowed ₹8000 at an annual rate of 4% compound interest. What will be the compound interest after 2 years if the interest is compounded annually.

(a) ₹ 652.8 (b) ₹ 8,652.8
(c) ₹ 8,625.8 (d) ₹ 625.8

RRB NTPC 07.04.2016 Shift : 2

Ans : (a)

$$\text{Compound interest} = \text{Principal}\left[\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}} - 1\right]$$

$$= 8000\left[\left(1+\frac{4}{100}\right)^2 - 1\right]$$

$$= 8000\left[\left(\frac{26}{25}\right)^2 - 1\right]$$

$$= 8000\left[\frac{676-625}{625}\right]$$

$$= 8000 \times \frac{51}{625}$$

$$= \frac{64 \times 51}{5}$$

$$= \frac{3264}{5}$$

= ₹ 652.8

86. Mr. Ayush borrowed ₹3000 at the rate of 5% compound interest annually. What will be the compound interest after 2 years?

(a) ₹370.5 (b) ₹307.5
(c) ₹3307.5 (d) ₹3370.5

RRB NTPC 19.01.2017 Shift : 1

Ans : (b) $C.I. = P\left(1+\frac{r}{100}\right)^n - P$

$$C.I. = 3000\left(1+\frac{5}{100}\right)^2 - 3000$$

$$= 3000\left[\left(\frac{21}{20}\right)^2 - 1\right]$$

$$= 3000\left(\frac{441}{400} - 1\right)$$

$$= 3000 \times \frac{41}{400}$$

$$= 7.5 \times 41 = 307.5$$

87. Mr. Yudish borrowed ₹3500 at an annual rate of 4% compound interest. What will be the compound interest for 2 years?

(a) ₹ 285.6 (b) ₹ 3785.6
(c) ₹ 3758.6 (d) ₹ 258.6

RRB NTPC 22.04.2016 Shift : 2

Ans : (a) P = ₹3500, r = 4 %
t = 2 Year, A = ?

$$A = P\left(1+\frac{r}{100}\right)^t$$

$$A = 3500\left(1+\frac{4}{100}\right)^2 = 3500\left(1+\frac{1}{25}\right)^2$$

$$A = 3500\left(\frac{26}{25}\right)^2$$

$$A = 3500\times\frac{26}{25}\times\frac{26}{25}$$

= 3785.6 रु.

interest = Amount – Principal
= 3785.6 – 3500 = ₹285.6

88. Mr. Ravish invests in a FD. What is the total amount he will get at maturity if ₹5500 is invested for 6 months at a compound interest rate of 20% per annum the interest is compounded quarterly?

(a) ₹6063.5 (b) ₹6063.75
(c) ₹6063.35 (d) ₹6063

RRB NTPC 26.04.2016 Shift : 3

Ans : (b) the interest is compounded quarterly,

Time $= \frac{6}{12}\times 4 = 2$ quarter

Rate $\frac{20}{4} = 5\%$

$$A = P\left(1+\frac{r}{100}\right)^n$$

$$A = 5500\left(1+\frac{5}{100}\right)^2$$

$$= 5500\left(1+\frac{1}{20}\right)^2$$

$$= 5500\times\frac{21}{20}\times\frac{21}{20}$$

$$= \frac{55\times 441}{4} = 6063.75$$

89. Mr. Manjunath borrowed ₹3500 at compound interest at an annual rate of 5%. What will be the compound interest of 2 years, if the interest is compounded annually?

(a) ₹358.75 (b) ₹3858.75
(c) ₹3885.75 (d) ₹385.75

RRB NTPC 26.04.2016 Shift : 3

Ans : (a) $A = P\left(1+\frac{r}{100}\right)^n = 3500\left(1+\frac{5}{100}\right)^2$

$$= 3500\times\frac{21}{20}\times\frac{21}{20} = \frac{35\times 441}{4} = \frac{15435}{4}$$

C.I. = Amount – Principal

$$CI = \frac{15435}{4} - 3500$$

$$= \frac{15435-14000}{4}$$

$$= \frac{1435}{4} = 358.75$$

90. Mr. Pritosh borrowed ₹4,500 at the rate of 5 percent compound interest. If the interest is compounded annually, what will be the compound interest for 2 years?

(a) ₹4961.25 (b) ₹461.25
(c) ₹4916.25 (d) ₹416.25

RRB NTPC 30.04.2016 Shift : 3

Ans : (b) P = 4500, r = 5%, t = 2 years, CI = ?

$$A = P\left(1+\frac{r}{100}\right)^t$$

$$A = 4500\left(1+\frac{5}{100}\right)^2$$

$$= 4500\left(1+\frac{1}{20}\right)^2$$

$$= 4500\left(\frac{21}{20}\right)^2$$

$$= 4500\times\frac{21}{20}\times\frac{21}{20}$$

= 4961.25
CI = A–P
CI = 4961.25 – 4500 = ₹461.25

91. The simple interest on a certain amount at 12% per annum for 3 years is ₹ 4,140. What is the compound interest on the same amount at 8% for two years?

(a) ₹2,012.40 (b) ₹1,840
(c) ₹1,913.60 (d) ₹1,886.50

RRB ALP & Tec. (29-08-18 Shift-III)

Ans : (c) $\because$ $SI = \frac{P\times R\times t}{100}$

$$4140 = \frac{P\times 12\times 3}{100}$$

P = 115×100
P = 11500

Again $\because$ $CI = P\left[\left(1+\frac{R}{100}\right)^t - 1\right]$

$$CI = 11500\left[\left(1+\frac{8}{100}\right)^2 - 1\right]$$

$$=11500\left[\left(1+\frac{2}{25}\right)^2-1\right]$$

$$=11500\left[\left(\frac{27}{25}\right)^2-1\right]$$

$$=11500\left[\frac{729-625}{625}\right]$$

$$=11500\left[\frac{104}{625}\right]=₹1913.60$$

92. Find the compound interest on ₹ 62500 at compound interest of 21% per annum for $1\frac{1}{2}$ years?

(a) ₹20,687.5 (b) ₹19,687.5
(c) ₹21,638.5 (d) ₹20,695

RRB ALP & Tec. (13-08-18 Shift-III)

Ans : (a) Given –

Principal = ₹62500

Rate of interest = 21% Annual

Time $=\frac{3}{2}$ Year

Then Compound amount= $62500\left(1+\frac{21}{100}\right)^{\frac{3}{2}}$

$$= 62500\left(\frac{121}{100}\right)^{\frac{3}{2}}$$

$$=62500\left(\frac{11}{10}\right)^3$$

Compound amount $=\frac{62500\times11\times11\times11}{1000}$

$$=\frac{625\times121\times11}{10}=₹83187.5$$

Now Compound interest

= 83187.5 – 62500 = ₹20687.5

Type - 4

93. The simple interest for 3 years at the rate of 8% on the amount of ₹'x' is equal to half of the compound interest of 2 years at the rate of 10% on the amount of ₹4000. Find the value of 'x'.

(a) ₹ 1750 (b) ₹ 1520
(c) ₹ 6000 (d) ₹ 1400

RRB JE - 28/06/2019 (Shift-III)

Ans. (a) Compound interest = Amount – Principal

$$= P\left(1+\frac{r}{100}\right)^n - P$$

$$= 4000\left(1+\frac{10}{100}\right)^2-4000$$

$$= 4000\times\frac{121}{100}-4000$$

$$= 4000\times\frac{21}{100}=840$$

As per the question–

Simple interest $=\frac{\text{compound interest}}{2}$

$\Rightarrow \frac{x\times8\times3}{100}=\frac{840}{2}$

$\Rightarrow \frac{x\times8\times3}{100}=420$

$\Rightarrow x=\frac{420\times100}{24}=\frac{7000}{4}=₹1750$

94. If the compound interest on a principal in 2 years at 5% per annum is ₹10 more than the simple interest on the same principal at the same time find the principal.

(a) ₹5,000 (b) ₹4,500
(c) ₹4,000 (d) ₹3,500

RRB RPF SI – 12/01/2019 (Shift-II)

Ans : (c) difference in compound interest and simple interest of two year

$$= \text{Principal}\left(\frac{r}{100}\right)^2$$

$$10 = \text{Principal}\left(\frac{5}{100}\right)^2$$ [∵ different = 10

r = 5%
n = 2

$$10 = \text{Principal}\left(\frac{1}{20}\right)^2$$

Principal = 10 × 4 00 = ₹4000

95. On a fixed amount, the compound interest and simple interest of 2 years at any interest rate is ₹696.30 and ₹660. Find the amount.

(a) ₹ 3000 (b) ₹ 4000
(c) ₹ 3300 (d) ₹ 3600

RRB JE - 29/05/2019 (Shift-I)

Ans : (a) Given -
C.I = ₹ 696.30
S.I. = ₹ 660
T = 2 Year
P = ?

Simple interest $=\frac{P\times R\times T}{100}$

$660 = \frac{P \times R \times 2}{100}$

$\frac{PR}{100} = 330$ — (i)

Compound interest $= P\left[\left(1+\frac{R}{100}\right)^2 - 1\right]$

$\Rightarrow 696.30 = P\left[1+\frac{R^2}{100}+\frac{2R}{100}-1\right]$

$\Rightarrow 696.30 = \frac{PR^2}{(100)^2}+\frac{2PR}{100}$

from equation (i)– $\frac{P \times R \times T}{100}$

$696.30 = \frac{330R}{100}+660$

$\Rightarrow 696.30 - 660 = \frac{330R}{100}$

$36.30 = \frac{330R}{100}$

$R = \frac{363}{33}$

$R = 11\%$

putting the value of R in equation (i)–

$\frac{PR}{100} = 330$

$P \times \frac{11}{100} = 330$

$P = \frac{330 \times 100}{11}$

$P = 30 \times 100$

P = ₹ 3000

So Principal will be ₹ 3000

96. The simple interest of 3 years at a 5% annual rate on a fixed amount is ₹1260. Find the compound interest on the same amount for the same period.

(a) ₹1324.05 (b) ₹1384.05
(c) ₹1428.05 (d) ₹1448

RRB RPF Constable – 22/01/2019 (Shift-III)

Ans : (a) Give–

Rate of simple interest (r) = 5%

time (t) = 3 Year

simple interest (S.I.) = ₹ 1260

simple interest $= \frac{P \times R \times T}{100}$

$1260 = \frac{P \times 5 \times 3}{100}$

$P = \frac{126000}{15}$

P = ₹ 8400

Compound interest = Principal $\left[\left(1+\frac{R}{100}\right)^t - 1\right]$

$= 8400\left[\left(1+\frac{5}{100}\right)^3 - 1\right]$

$= 8400\left[\left(\frac{21}{20}\right)^3 - 1\right]$

$= 8400\left[\frac{9261-8000}{8000}\right]$

$= 8400 \times \frac{1261}{8000}$

$= \frac{84 \times 1261}{80}$

= ₹ 1324.05

97. The compound interest of 2 years at a rate of 10% on a fixed amount is ₹2100. What will be the simple interest on the same amount at the same rate for the same period?

(a) ₹1600 (b) ₹1980
(c) ₹1800 (d) ₹2000

RRB JE - 26/05/2019 (Shift-II)

Ans : (d) compound interest $= P\left[\left(1+\frac{r}{100}\right)^n - 1\right]$

$2100 = P\left[\left(1+\frac{10}{100}\right)^2 - 1\right]$

$2100 = P\left[\frac{121}{100} - \frac{1}{1}\right]$

$2100 = P\left[\frac{121-100}{100}\right]$

$2100 = P \times \frac{21}{100}$

$P = \frac{2100 \times 100}{21}$

Principal (P) = 10000

As per the question,

simple interest $= \frac{P \times R \times T}{100}$

$SI = \frac{10000 \times 10 \times 2}{100}$

$\boxed{SI = 2000}$

So SI will be ₹2000.

98. An amount was deposited for 7 years at 8% simple interest rate. The mature amount was invested in a scheme with an annual rate of 10% compound interest, yielding an interest of ₹1638 over 2 years. Find the principal amount.

(a) ₹6200 (b) ₹5000
(c) ₹7500 (d) ₹8000

RRB JE - 27/05/2019 (Shift-I)

Ans : (b) Let Principal Amount = ₹ P

simple interest = $P \times \frac{8}{100} \times 7 = \frac{56P}{100}$

Amount = $P + \frac{56P}{100} = \frac{156P}{100}$

As per the question,

Compound interest = ₹ 1638

$1638 = \frac{156P}{100}\left[\left(1+\frac{10}{100}\right)^2 - 1\right]$

$\frac{1638 \times 100}{156P} = \frac{21}{100}$

$P = \frac{1638 \times 100 \times 100}{156 \times 21}$

P = ₹ 5000

99. On an amount, the simple interest of 2 years is ₹660, while the compound interest of two years is ₹696.30, keeping the rate of interest same. Find the rate of interest?

(a) 13% (b) 11%
(c) 10% (d) 12.75%

RRB JE - 27/05/2019 (Shift-II)

Ans : (b) Simple interest $(SI) = \frac{PRT}{100}$

$660 = \frac{P \times R \times 2}{100}$

PR = 330 × 100

PR = 33000 ..... (i)

difference in compound interest and simple interest for two year.

$= P\left(\frac{R}{100}\right)^2$

$696.30 - 660 = P \times \frac{R}{100} \times \frac{R}{100}$

$36.30 = \frac{33000 \times R}{100 \times 100}$ [PR = 33000

36.30 = 3.3 × R from the eq^n (1)]

$R = \frac{363}{33}$

$R = \frac{33}{3}$

R = 11%

100. An amount at a fixed rate of simple interest increases by 100% in 8 years What will be the compound interest for 2 years for ₹8000 at the same rate?

(a) ₹2050 (b) ₹2075
(c) ₹2125 (d) ₹2025

RRB JE - 29/05/2019 (Shift-III)

Ans : (c) Let Principal = P

Amount = 2P

interest = 2P – P = P

simple interest = $\frac{P \times R \times T}{100}$

$P = \frac{P \times R \times 8}{100}$

$R = \frac{25}{2}\%$

from question –

Compound interest $= P\left[\left(1+\frac{Rate}{100}\right)^{Time} - 1\right]$

$= 8000\left[\left(1+\frac{25}{200}\right)^2 - 1\right] = 8000 \times \frac{(81-64)}{64}$

$= 8000 \times \frac{17}{64} =$ ₹ 2125

101. Simple interest and compound interest on a fixed amount for two years are ₹8400 and ₹8652 respectively. Find the rate of interest.

(a) 5% (b) 6%
(c) 4.5% (d) 5.5%

RRB JE - 29/05/2019 (Shift-III)

Ans : (b) When time is 2 year then–

$C.I. - S.I. = P\left(\frac{R}{100}\right)^2$

$8652 - 8400 = \frac{PR^2}{10000}$

$252 = \frac{PR^2}{10000}$.....(i)

$S.I. = \frac{P \times R \times 2}{100}$

$8400 = \frac{PR}{50}$

PR = 8400 × 50 ..... (ii)

from the equation (i) and (ii)–

$PR^2 = 252 \times 10000$

PR × R = 252 × 10000

$R = \frac{252 \times 10000}{PR}$

$R = \frac{252 \times 10000}{8400 \times 50}$

$R = \frac{21 \times 10}{7 \times 5}$

R = 6%

102. Find the simple interest of same amount for, 2 years at the rate of 12.5% on an amount, if the compound interest of the same period at the same rate is ₹510?

(a) ₹480 (b) ₹500
(c) ₹408 (d) ₹420

RRB JE - 30/05/2019 (Shift-III)

Ans : (a) Compound interest = Amount – Principal

$$510 = P\left[1+\frac{r}{100}\right]^n - P$$

$$= P\left[1+\frac{r}{100}\right]^n - P$$

$$= P\left[1+\frac{12.5}{100}\right]^2 - P$$

$$= P\left[\frac{1125}{1000}\right]^2 - P$$

$$= P\left[\frac{9}{8}\right]^2 - P \quad = P\left[\frac{81-64}{64}\right]$$

$$510 = \frac{P\times 17}{64}$$

$$P = \frac{510\times 64}{17}$$

P = 1920

$$\text{Simple interest} = \frac{P\times R\times T}{100}$$

$$\text{Simple interest} = \frac{1920\times 12.5\times 2}{100}$$

$$\text{Simple interest} = \frac{1920\times 25}{100}$$

Simple interest = ₹480

103. Simple interest on a certain sum of money for 3 years at 5% per annum is ₹ 5250. What will be the compound interest on that sum of money at the same rate for the same period?

(a) ₹5,510.88 (b) ₹5,516.88
(c) ₹5,512.88 (d) ₹5,517.88

RRB ALP & Tec. (10-08-18 Shift-II)

Ans : (b) simple interest $= \frac{P\times R\times T}{100}$

$$5250 = \frac{P\times 5\times 3}{100}$$

$$\text{Principal} = \frac{5250\times 20}{3}$$

Principal = 35000

$$\text{Compound interest} = \text{Principal}\left[\left(1+\frac{r}{100}\right)^n - 1\right]$$

$$= 35000\left[\left(1+\frac{5}{100}\right)^3 - 1\right]$$

$$= 35000\left[\frac{21\times 21\times 21 - 20\times 20\times 20}{20\times 20\times 20}\right]$$

$$= 35000\left(\frac{9261-8000}{8000}\right)$$

$$= 35\times\frac{1261}{8}$$

= 35 × 157.625 = ₹5516.88

104. Smita took a loan of ₹75000 at a simple annual interest rate of 15%. She give that money to her friend on the same day, at the same interest rate but according to the annual compound interest rate. How much profit will she earn after 2 years?

(a) ₹ 1,887.5 (b) ₹ 1,867.5
(c) ₹ 1,786.5 (d) ₹ 1,687.5

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (d) simple interest (SI) $= \frac{P\times R\times T}{100}$

$$= \frac{75000\times 15\times 2}{100} = ₹\ 22500$$

Compound amount

$$A = P\left(1+\frac{R}{100}\right)^n$$

$$A = 75000 \times \frac{115}{100}\times\frac{115}{100}$$

A = ₹ 99187.5

Compound interest = A – P

= 99187.5 – 75000 = ₹24187.5

Smita's Profit = 24187.5 – 22500 = ₹1687.5

105. Anjali borrowed ₹22500 for 2 years at the rate of 6% simple annual interest and on the same day she gives this amount to Ashoka at compound interest for the same interest rate and same time. How much loss or profit will she make?

(a) 73 Profit (b) 73 Loss
(c) 81 Loss (d) 81 Profit

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (d) R = 6%

T = 2

P = 22500

$$S.I. = \frac{22500\times 6\times 2}{100}$$

S.I. = 2700

= 22500 + 2700

Amount = 25200

$$A = 22500\left(1+\frac{6}{100}\right)^2 = 22500\times\frac{53}{50}\times\frac{53}{50}$$

= 25281

Total Profit = 25281 – 25200 = ₹81

106. ₹100 has been invested in a scheme for one year in which annual interest is given at the rate of 10% Another ₹100 has been invested in a one year plan that yields an annual interest rate of 10% but the interest is compounded half yearly The interest received by second plan is how much more than the first plan.

(a) ₹ 1 (b) 50 Paisa
(c) 25 Paisa (d) No difference

RRB Group-D – 16/11/2018 (Shift-III)

Ans. (c) : Received simple interest of invested amount in first scheme $= \frac{100\times10\times1}{100}$ =₹ 10

Received simple interest of invested amount in second scheme

$$=100\left(1+\frac{5}{100}\right)^2-100$$

$$=100\times\left(\frac{21}{20}\right)^2-100$$

$$=100\times\frac{441}{400}-100$$

$$=\frac{441}{4}-100\Rightarrow\frac{441-400}{4}=\frac{41}{4}=₹10.25$$

Required difference = 10.25 – 10 = 25 paisa

107. A simple interest of 8% on an amount for 1.5 year is ₹360. If the time is doubled according to the same interest rate on the same amount, find the compound interest?

(a) ₹ 778.13 (b) ₹ 779.13
(c) ₹ 779 (d) ₹ 778

RRB Group-D – 01/10/2018 (Shift-III)

Ans : (b) As per the question,

$$\text{Simple interest} = \frac{Prt}{100}$$

$$360=\frac{P\times1.5\times8}{100}$$

P = ₹3000

doubling the time (t) = 2×1.5 year = 3 year

$$\text{Compound interest} = P\left[\left(1+\frac{r}{100}\right)^t-1\right]$$

$$=3000\left[\left(1+\frac{8}{100}\right)^3-1\right]$$

$$=3000\left[\left(1+\frac{2}{25}\right)^3-1\right]$$

$$=3000\left[\left(\frac{27}{25}\right)^3-1\right]=3000\left[\frac{19683}{15625}-1\right]$$

$$=3000\left[\frac{19683-15625}{625\times25}\right]$$

$$=\frac{4058\times3000}{625\times25}=\frac{4.8\times4058}{25}=\frac{19478.4}{25}$$

= ₹ 779.13

108. The simple interest of 8 years at 8% per annum on a deposit is ₹16000.What will be the compound interest of 2 years at the rate of one fourth of this rate on the same amount.

(a) ₹ 1,020 (b) ₹ 980
(c) ₹ 1,010 (d) ₹ 1,015

RRB NTPC 17.01.2017 Shift-2

Ans : (c) simple interest = $\frac{P\times R\times T}{100}$

$$\text{Principal}=\frac{16000\times100}{8\times8}=25000$$

As per the question,

$$r = 8\%\ \text{का}\ \times\frac{1}{4}$$

$$\text{Compound interest} = 25000\left(1+\frac{8\times\frac{1}{4}}{100}\right)^2-25000$$

$$=\frac{25000\times51\times51}{50\times50}-25000=26010-25000=1010$$

So Compound interest = ₹1010

109. Simple interest on a fixed sum of money is ₹1200 over 2 years at a fixed rate of interest. The compound interest of the same amount gets ₹1290 in 2 years at the rate of simple interest. What will be the principle?

(a) ₹1200 (b) ₹16000
(c) ₹6000 (d) ₹4000

RRB NTPC 02.04.2016 Shift : 2

Ans : (d) Rate = R, Time (T) = 2 years

SI = 1200, CI = 1290

and difference of CI to SI of two years

सूत्र $\boxed{D=P\left(\frac{R}{100}\right)^2}$

$$1290-1200=P\left(\frac{R^2}{10000}\right)$$

PR^2 = 90 × 10000

PR^2 = 900000 ……..(1)

$$SI=\frac{PTR}{100}$$

$$1200=\frac{P\times2\times R}{100}$$

$$PR=\frac{1200\times100}{2}$$

PR = 60000 …….. (2)

From the equation (1)–

PR.R = 900000

60000 × R = 900000

$$R=\frac{900000}{60000}$$

R = 15%

From the equation (2)–

P × 15 = 60,000

$$P=\frac{60000}{15}$$ (Principal) P = ₹4000

110. An interest of ₹2000 is given on a fixed sum of money at a rate of 10% per annum simple interest in 2 years. If the compound interest accrues annually on this amount, what will be the effective rate of interest?

(a) 10.25 (b) 10.50
(c) 10.75 (d) 10.15

RRB NTPC 30.03.2016 Shift : 2

Ans : (b) Let Amount = x

Then,

$$2000 = x \times \frac{10}{100} \times 2$$

$$x = ₹10,000$$

$$CI = 10000\left(1+\frac{10}{100}\right)^2 - 10000$$

$$= 10000\left(\frac{121}{100} - 1\right) = 2100$$

So now interest rate on ₹2100

$$r = \frac{2100 \times 100}{10000 \times 2} = \frac{21}{2} = 10.5\%$$

111. On certain principal, simple interest is obtained to ₹ 2,000 in two years at the rate of 10% p.a. if the interest on this amount is compounded annually. So what will be the difference between the two types of interest?

(a) ₹200 (b) ₹220
(c) ₹100 (d) ₹120

RRB NTPC 29.03.2016 Shift : 3

Ans : (c) Let Principal = ₹x

$\because$ Simple interest $= \frac{P \times R \times T}{100}$

$$\Rightarrow 2000 = \frac{x \times 10 \times 2}{100}$$

$$x = 10000$$

$\therefore$ Compound interest for 2 years $= P\left[\left(1+\frac{r}{100}\right)^n - 1\right]$

$$= 10000\left[\left(1+\frac{10}{100}\right)^2 - 1\right]$$

$$= 10000\left[\frac{121}{100} - 1\right]$$

$$= 10000 \times \frac{21}{100} = ₹2100$$

$\therefore$ Different in interest = 2100 – 2000 = ₹100

112. There was an increase of 40% by the simple interest on an amount in 4 years. What will be the compound interest at ₹6000 at the same rate in 3 years?

(a) ₹ 1260 (b) ₹ 1886
(c) ₹ 1986 (d) ₹ 7986

RRB NTPC 18.01.2017 Shift : 3

Ans : (c)

Let rate of interest = r %

Principal = ₹100

$$\text{interest} = 100 \times \frac{40}{100} = 40$$

As per the question

$$\Rightarrow 40 = \frac{100 \times 4r}{100}, r = 10\%$$

Compound interest of 3 years of ₹6000

$$\Rightarrow 6000 \times \left[\left(1+\frac{10}{100}\right)^3 - 1\right] = ₹1986$$

113. If the rate of interest is 8% per annum and ₹10000 are lent for half yearly at the compound interest, then calculate the rate of equivalent simple interest for the first year?

(a) 8.16% (b) 9%
(c) 7% (d) 10%

RRB NTPC 19.04.2016 Shift : 1

Ans : (a) $\because$ Interest is compounded for half yearly.

$$\therefore r = \frac{8}{2} = 4\% \text{ half yearly}$$

n = 1 year = 1 × 2 = 2 half yearly

Compound interest for the year

$$CI = P\left[\left(1+\frac{r}{100}\right)^n - 1\right]$$

$$= 10000\left[\left(1+\frac{4}{100}\right)^2 - 1\right]$$

$$= 10000\left[\frac{26}{25} \times \frac{26}{25} - 1\right]$$

$$= 10000\left[\frac{676-625}{625}\right]$$

$$= 10000 \times \frac{51}{625}$$

= ₹816

If simple interest be ₹816 for one year.

$$\text{then } 816 = \frac{10000 \times R \times 1}{100}$$

$$R = 8.16\%$$

$\therefore$ equivalent simple interest rate = 8.16%

114. Find the compound interest at 7% of that amount for 3 years where whose simple interest will be ₹18900 at the rate of 7% in 3 years. (Rounded off to nearest ₹)

(a) ₹19,746 (b) ₹18,390
(c) ₹20,254 (d) ₹21,053

RRB NTPC 28.04.2016 Shift : 1

Ans : (c) simple interest (S.I) $= \frac{P \times R \times T}{100}$

$$18,900 = \frac{P \times 7 \times 3}{100}$$

$$P = \frac{18,900 \times 100}{7 \times 3} = 900 \times 100 = 90000$$

Again compound interest

$$(C.I) = 90000\left[\left(1+\frac{7}{100}\right)^3 - 1\right]$$

$$= 90000\left[\left(\frac{107}{100}\right)^3 - 1\right]$$

$$= 90000\left(\frac{1225043}{1000000} - 1\right)$$

$$= 90000 \times \frac{225043}{1000000} = 9 \times \frac{225043}{100}$$

$$= 20253.87 = ₹20254$$

Type - 5

115. The difference of 2 years compound interest and simple interest on a fixed amount at 12.5% is ₹45. Find the amount.

(a) ₹ 2880 (b) ₹ 3000
(c) ₹ 2000 (d) ₹ 2440

RRB JE - 22/05/2019 (Shift-I)

Ans : (a) r = 12.5% , t = 2 Year CI ~ SI = ₹45 P = ?
difference between CI and SI of two years

$$= P \times \left(\frac{r}{100}\right)^2$$

$$45 = P \times \frac{(12.5)^2}{(100)^2}$$

$$P = \frac{45 \times 100 \times 100}{12.5 \times 12.5} = ₹\ 2880$$

116. At a fixed amount for, 2 years simple interest at 4% is ₹140. Find the difference between the compound interest and the simple interest for the same rate and the same period at the same amount.

(a) ₹ 2.80 (b) ₹ 2.40
(c) ₹ 3 (d) ₹ 1.80

RRB JE - 23/05/2019 (Shift-I)

Ans : (a) Given–
R = 4% time = 2 Year
Simple interest = 140

Simple interest $= \frac{P \times R \times T}{100}$

$$140 = \frac{P \times 4 \times 2}{100}$$

$$P = \frac{7000}{4} = 1750$$

As per the question –

$$A = P\left(1+\frac{r}{100}\right)^n$$

$$A = 1750\left(1+\frac{4}{100}\right)^2$$

$$A = 1750 \times \frac{26}{25} \times \frac{26}{25}$$

$$A = 1892.8$$

Interest = 1892.8 – 1750 = 142.8
Difference of interest = 142.8 – 140 = ₹ 2.8

117. Find the difference between compound interest and simple interest for 2 years at the rate of 8% per annum on ₹15000.

(a) ₹100 (b) ₹96
(c) ₹108 (d) ₹12

RRB RPF SI – 13/01/2019 (Shift-III)

Ans : (b) Given-
P = ₹ 15000
R = 8%
time (t) = 2 Year

When time is 2 year and then difference of compund interest and simple interest.

$$D = P\left(\frac{R}{100}\right)^2 \text{ from,}$$

$$= 15000\left(\frac{8}{100}\right)^2$$

$$= 15000 \times \frac{4}{625} = ₹\ 96$$

118. On what principal will the difference of compound interest and simple interest at the rate of 10% for three years will be ₹620?

(a) ₹ 18000 (b) ₹ 25000
(c) ₹ 24000 (d) ₹ 20000

RRB JE - 24/05/2019 (Shift-III)

Ans : (d) Let Principal = ₹P
As per the question–

$$620 = \left[P\left(1+\frac{10}{100}\right)^3 - 1\right] - \frac{P \times 10 \times 3}{100}$$

$$620 = P\left[\left(\frac{1331-1000}{1000}\right) - \frac{10 \times 3}{100}\right]$$

$$620 = P\left[\frac{331-300}{1000}\right]$$

$$620 = P\left[\frac{31}{1000}\right]$$

P = ₹20000

119. Find the difference between compound interest and simple interest, received in 4 years at the rate of 10% on the amount of ₹ 1000.

(a) ₹64.10 (b) ₹52
(c) ₹74 (d) ₹16.40

RRB JE - 26/05/2019 (Shift-III)

Ans. (a)

Compound amount after n Year $A = P\left(1+\frac{r}{100}\right)^n$

Compound amount after 4 Year $=1000\left(1+\frac{10}{100}\right)^4$

$=1000\times\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10}$ = ₹1464.1

∴ Compound interest = 1464.1 – 1000 = 464.1

Simple interest after 4 years $=\frac{PRT}{100}$

$=\frac{1000\times10\times4}{100}=400$

So intended difference = 464.1 – 400 = ₹64.10

120. The difference between compound interest and simple interest of 2 years at a rate of 8% on a fixed amount is ₹8. Find the amount.

(a) ₹2000 (b) ₹1250
(c) ₹1500 (d) ₹1000

RRB JE - 02/06/2019 (Shift-I)

Ans : (b) Given–

interest Rate = 8%

time = 2 Year

difference of compound interest and simple interest= ₹8

Let amount = P [where D = difference]

$D=P\left(\frac{R}{100}\right)^2$ from–

$8=P\left(\frac{8}{100}\right)^2$

$8=P\times\frac{4}{625}$

P = ₹1250

121. The difference between compound interest and simple interest is ₹124.05 when calculated on a half yearly basis for 2 years at the rate of 10% on a fixed amount. Find the amount.

(a) ₹ 8400 (b) ₹ 10000
(c) ₹ 8000 (d) ₹ 8200

RRB JE - 02/06/2019 (Shift-I)

Ans : (c) Given

interest Rate = R/2 = 10/2 = 5% (Half yearly)

time = 2 Year = 4 Half year

Let Principal = P

$P\left[\left(1+\frac{R}{100}\right)^T-1\right]-\left(\frac{P\times R\times T}{100}\right)=124.05$

$P\left[\left(1+\frac{5}{100}\right)^4-1\right]-\left(\frac{P\times5\times4}{100}\right)=124.05$

$P\left[\frac{21\times21\times21\times21-20\times20\times20\times20}{20\times20\times20\times20}\right]-\left(\frac{P\times5\times4}{100}\right)=124.05$

$P\left[\frac{194481-160000}{160000}\right]-\frac{P\times5\times4}{100}=\frac{12405}{100}$

$P\left[\frac{34481-32000}{160000}\right]=\frac{12405}{100}$

$P\left[\frac{2481}{1600}\right]=12405$

$P=\frac{12405\times1600}{2481}$ = ₹8000

122. The difference between the simple interest on a principal, at the rate of 8% in 8 months and at the rate of 6% in 14 months is 200. Find the principal.

(a) ₹12000 (b) ₹12800
(c) ₹24000 (d) ₹14400

RRB JE - 01/06/2019 (Shift-I)

Ans : (a) Let Principal = ₹x

simple interest of 8 month at the rate of 8%

Rate (R) = 8%, time $(T)=\frac{8}{12}$ Year $=\frac{2}{3}$ Year

simple interest $=\frac{P\times R\times T}{100}$

$=\frac{x\times8\times\frac{2}{3}}{100}$

$=\frac{16x}{300}$

simple interest of 14 months at the rate of 6%

Rate (R) = 6%, time $(T)=\frac{14}{12}=\frac{7}{6}$ Year

simple interest $=\frac{P\times R\times T}{100}$

$=\frac{x\times6\times\frac{7}{6}}{100}$

$=\frac{7x}{100}$

difference between simple interest = 200

$\frac{7x}{100}-\frac{16x}{300}=200$

$\frac{21x-16x}{300}=200$

$\frac{5x}{300}=200$

x = 40 × 300

x = ₹12000

123. The difference between simple interest received on a sum of money at 4% per annum for 2 years and compound interest is ₹8. Find that amount.

(a) ₹4000 (b) ₹10000
(c) ₹8000 (d) ₹5000

RRB JE - 01/06/2019 (Shift-III)

Ans. (d) for 2 Year,
Formula:– Compound interest – simple interest

$$= \text{Principal}\left(\frac{\text{Rate}}{100}\right)^{\text{time}}$$

$$8 = \text{Principal}\left(\frac{4}{100}\right)^2$$

$$8 = \text{Principal} \times \frac{1}{25} \times \frac{1}{25}$$

Principal = 8 × 25 × 25
Principal = ₹5000

124. The difference between simple interest and compound interest on a certain amount at the rate of 5% for 3 years is ₹14.48. Find the amount.
(a) ₹ 1850 (b) ₹ 1999
(c) ₹ 1899 (d) ₹ 2160
RRB RPF Constable – 22/01/2019 (Shift-III)

Ans : (c) simple interest $= \frac{P \times R \times T}{100}$

Let P = ₹x

simple interest $= \frac{x \times 5 \times 3}{100} = \frac{3x}{20}$

Compound interest $= x\left[\left(1+\frac{5}{100}\right)^3 - 1\right]$

$$= x\left[\left(1+\frac{1}{20}\right)^3 - 1\right] = x\left[\left(\frac{21}{20}\right)^3 - 1\right]$$

$$= x\left(\frac{9261-8000}{8000}\right) = \frac{1261}{8000}x$$

As per the question,

$$\frac{1261}{8000}x - \frac{3x}{20} = 14.48$$

$$\frac{1261x - 1200x}{8000} = 14.48$$

$$61x = 14.48 \times 8000$$

$$x = \frac{1448 \times 80}{61} = ₹1899$$

125. Find the difference between compound interest and simple interest of ₹400 for 2 years at an interest rate of 5%.
(a) ₹10 (b) ₹4 (c) ₹3 (d) ₹1
RRB RPF Constable – 24/01/2019 (Shift-III)

Ans. (d) When difference is two year between simple interest and compound interest,

$$\text{Difference} = \text{Principal}\left(\frac{R}{100}\right)^2$$

$$\text{Difference} = P\left(\frac{r}{100}\right)^2$$

r = 5%, P = ₹400

$$\text{Difference} = 400\left(\frac{5}{100}\right)^2 = \frac{400}{20 \times 20} = ₹1$$

126. Find the difference between simple interest and compound interest at a ₹2000 for 2 years at 10% annual interest rate.
(a) ₹ 0 (b) ₹ 30
(c) ₹ 20 (d) ₹ 10
RRB Group-D – 25/09/2018 (Shift-III)

Ans. (c) : Given-
t = 2 Year
p = ₹ 2000
r = 10%
Formula-

$$D = P\left(\frac{R}{100}\right)^2$$

$$D = 2000\left(\frac{10}{100}\right)^2$$

$$D = 2000 \times \frac{1}{100}$$

D = ₹20

127. The difference between compound interest and simple interest is ₹2 for 2 years at an interest rate of 5% on an amount. Find the principal amount.
(a) ₹650 (b) ₹920
(c) ₹800 (d) ₹700
RRB Group-D – 29/10/2018 (Shift-III)

Ans : (c)

$D = P\left(\frac{R}{100}\right)^2$ from the formula,

$$2 = P\left(\frac{5}{100}\right)^2$$

P = 2 × 20 × 20 = ₹ 800

128. Find the difference between compound interest and simple interest at ₹8000 at the interest rate of 5% for 3 years
(a) ₹ 61 (b) ₹ 58
(c) ₹ 95 (d) ₹ 68
RRB Group-D – 30/10/2018 (Shift-II)

Ans : (a) $A = P\left(1+\frac{R}{100}\right)^n$

$$A = 8000\left(1+\frac{5}{100}\right)^3$$

$$A = 8000 \times \frac{105}{100} \times \frac{105}{100} \times \frac{105}{100}$$

A = ₹ 9261

Compound interest = Amount (A) – Principal (P)
= 9261 – 8000 = ₹ 1261

simple interest (SI) $= \frac{P \times R \times T}{100}$

$$= \frac{8000 \times 5 \times 3}{100} = 1200$$

difference between compound interest and simple interest = 1261 – 1200 = ₹ 61

129. The difference between simple interest and compound interest received in two years at ₹20,000 is ₹800. What is the annual rate of interest?

(a) 30% (b) 15%
(c) 25% (d) 20%

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (d) Difference of simple interest and compound interest of 2 year = ₹800

Formula $\boxed{d = \frac{PR^2}{100^2}}$

d = difference, P = Principal
R = Rate

$800 = \frac{20000 \times R^2}{100 \times 100}$

$R^2 = 400$

$\boxed{R = 20\%}$

130. These is a difference of ₹180 between 15% simple annual interest and compound annual interest on a certain amount for 2 years Find the principal amount.

(a) ₹ 9000 (b) ₹ 7000
(c) ₹ 8000 (d) ₹ 6000

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (c) Let Principal = ₹ P

simple interest = $\frac{P \times 15 \times 2}{100}$ ____(i)

Compound interest $= P\left[\left(1+\frac{15}{100}\right)^2 - 1\right]$

$= P\left[\left(\frac{23}{20}\right)^2 - 1\right]$

$= P\left[\frac{529-400}{400}\right] = \frac{P \times 129}{400}$ _____(ii)

As per the question-

Compound interest ~ simple interest = 180

$= \frac{129P}{400} - \frac{30P}{100} = 180$

$\frac{129P - 120P}{400} = 180$

9P = 180 × 400

P = 20 × 400

P = ₹8000

131. What will be the difference between simple interest and compound interest of 2 years at 10% interest rate on ₹8500?

(a) ₹ 68 (b) ₹ 50
(c) ₹ 85 (d) ₹ 70

RRB Group-D – 23/10/2018 (Shift-II)

Ans. (c) :
Difference of simple interest and Compound interest of 2 year is

$D = P\left(\frac{R}{100}\right)^2$

$D = \frac{8500 \times 10}{100} \times \frac{10}{100}$ = ₹ 85

132. The difference between the simple interest and compound interest of the amount invested at 5% per annum for 2 years is ₹15. Find the amount.

(a) ₹ 4,980 (b) ₹ 3,000
(c) ₹ 4,800 (d) ₹ 6,000

RRB Paramedical Exam – 21/07/2018 (Shift-III)

Ans : (d) Difference of simple interest and Compound interest of 2 year is–

$D = P\left(\frac{R}{100}\right)^2$

$15 = P\left(\frac{5}{100}\right)^2$

P = 15 × 20 × 20 = ₹6000

133. The difference in simple interest paid by two banks for two years at ₹ 5000 is ₹25. Explain the difference between the interest rates of the two banks.

(a) 0.25% (b) 0.60%
(c) 1% (d) 0.10%

RRB Group-D – 23/10/2018 (Shift-III)

Ans : (a) Suppose the first bank gives interest at the rate of R_1% and the second bank gives interest at the rate of R_2%.

Interest by first bank $= \frac{5000 \times R_1 \times 2}{100} = 100R_1$ .....(i)

Interest by second bank

$= \frac{5000 \times R_2 \times 2}{100} = 100R_2$ .....(ii)

As per the question,

$100\ R_1 - 100\ R_2 = 25$

$100\ (R_1 - R_2) = 25$

$R_1 - R_2 = \frac{25}{100} = \frac{1}{4} = 0.25\%$

134. The difference between the compound interest and simple interest of an amount at the rate of 15% per annum for 3 years is ₹283.50. Find the amount.

(a) ₹ 4,040 (b) ₹ 4,000
(c) ₹ 4,400 (d) ₹ 4,444

RRB Group-D – 12/10/2018 (Shift-I)

Ans. (b) :
Difference = Compound interest – simple interest

Difference = $\left[P\left(1+\frac{R}{100}\right)^3 - P\right] - \left(\frac{PRT}{100}\right)$

$$283.50 = P\left[\left(1+\frac{15}{100}\right)^3 - \frac{15\times 3}{100} - 1\right]$$

$$\Rightarrow \quad 283.50 = P\left[\frac{115}{100}\times\frac{115}{100}\times\frac{115}{100} - \frac{45}{100} - 1\right]$$

$$\Rightarrow \quad 283.50 = P\left[\frac{12{,}167 - 3600 - 8000}{8000}\right]$$

$$P = \frac{283.5\times 8000}{567}$$

$$P = ₹\,4000$$

135. Find the difference between simple interest and compound interest of 2 years at an annual rate of 20% on a principal of ₹4000.

(a) 160 (b) 120
(c) 90 (d) 110

RRB NTPC 09.04.2016 Shift : 3

Ans : (a) Principal (P) = ₹4000
Rate (R) = 20%
Difference of CI and SI of two years

Compound interest – simple interest $= P\times\left(\frac{R}{100}\right)^2$

$$= 4000\times\left(\frac{20}{100}\right)^2 = 4000\times\frac{20}{100}\times\frac{20}{100} = ₹160$$

136. Find the difference between the maturity values of (i) simple interest for 5 years and (ii) annual compound interest at the same rate for a amount of ₹5000.

(a) ₹11.00 (b) ₹11.50
(c) ₹12.00 (d) ₹12.50

RRB NTPC 06.04.2016 Shift : 2

Ans : (d) Principal = ₹5000
Rate = 5%
time = 2 Year
difference in Compound interest and simple interest for two years

$$= \left(\frac{R}{100}\right)^2 \times \text{ Principal } = \left(\frac{5}{100}\right)^2 \times 5000$$

$$= \frac{25}{100\times 100}\times 5000 = ₹12.50$$

137. Each of the two amounts of ₹10,000 was invested for 2 years (i) at the rate of 5% simple interest (ii) at the same rate of annual compound interest. Whjat is the difference in their moturity value?

(a) ₹ 30 (b) ₹ 25
(c) ₹ 20 (d) ₹ 40

RRB NTPC 26.04.2016 Shift : 1

Ans : (b)
Difference between of CI and SI for two **years**

$$(D) = P\left(\frac{R}{100}\right)^2 = 10000\left(\frac{5}{100}\right)^2$$

$$= 10000\times\frac{25}{10000} = ₹\,25$$

138. M borrowed some money for 1 year at certain simple interest. But the interest rate was increased to 2% then the difference of simple interest is ₹ 120. Find the principal amount.

(a) ₹4,000 (b) ₹5,000
(c) ₹6,000 (d) ₹7,000

RRB NTPC 27.04.2016 Shift : 2

Ans : (c) As per the question-

$$\frac{P\times(r+2)\times 1}{100} - \frac{P\times r\times 1}{100} = 120$$

$$\frac{P}{100}(r+2-r) = 120$$

$$\frac{P}{100}\times 2 = 120$$

$$P = 120\times 50$$

$$P = ₹6000$$

139. Find the difference between simple interest and compound interest of ₹23,465 received at the rate of 7.5% in the second year.

(a) 132 (b) 66
(c) 147 (d) 73.5

RRB NTPC 28.04.2016 Shift : 1

Ans : (a) Given, Principal (P) = 23,465
and Rate (r) = 7.5%
Formula for difference of 2 years compound interest and simple interest–

$$\text{Difference (D)} = P\left(\frac{r}{100}\right)^2 = 23{,}465\left(\frac{7.5}{100}\right)^2$$

$$= 23465\left(\frac{75}{1000}\right)^2 = 23465\left(\frac{3}{40}\right)^2$$

$$= 23465\times\frac{9}{1600} = \frac{211185}{1600} \Rightarrow 131.99 = 132$$

Type - 6

140. The amount lent at a fixed rate of compound interest becomes ₹1460 in 2 years and ₹1606 in 3 years Find the rate of interest.

(a) 11% (b) 12%
(c) 10% (d) 8%

RRB JE - 02/06/2019 (Shift-II)

Ans. (c) From $A = P\left(1+\frac{r}{100}\right)^n$

$$\frac{\text{Amount of 3 years}}{\text{Amount of 2 years}} = \frac{1606}{1460}$$

$$\frac{P\left(1+\frac{r}{100}\right)^3}{P\left(1+\frac{r}{100}\right)^2} = \frac{1606}{1460}$$

$$\frac{1606}{1460}=\left(1+\frac{R}{100}\right)^1$$

$$\frac{803}{730}=\left(1+\frac{R}{100}\right)^1$$

$$\frac{803}{730}-1=\frac{R}{100}$$

$$R=\frac{803-730}{730}=\frac{73}{730}\times 100=10\%$$

141. Deepa deposits an amount of ₹6250 in the bank which becomes ₹7840 at the rate of compound annual interest in two years the rate of interest is?

(a) 13% (b) 10%

(c) 12% (d) 11%

RRB Group-D – 25/09/2018 (Shift-I)

Ans : (c) Principal = ₹ 6250

Time = 2 Year

Amount = ₹ 7840

Let Rate = R % Annual

$$\text{Amount} = \text{Principal} \times\left(1+\frac{R}{100}\right)^2$$

$$7840=6250\times\left(1+\frac{R}{100}\right)^2$$

$$\frac{7840}{6250}=\left(\frac{100+r}{100}\right)^2$$

$$\left(\frac{28}{25}\right)^2=\left(\frac{100+R}{100}\right)^2$$

$$\Rightarrow\frac{28}{25}=\frac{100+R}{100}$$

$\Rightarrow 28 \times 4 = 100 + R$

R = 112 – 100

R = 12% Annual

142. Hari invested ₹100 for three years at a simple interest rate of 11.03%. How much should tipu invest to get the same amount after three years but at the rate of 10% compound interest?

(a) ₹ 120 (b) ₹ 110

(c) ₹ 100 (d) ₹ 105

RRB Group-D – 05/11/2018 (Shift-I)

Ans. (c) : As per the question–

$$\frac{11.03\times100\times3}{100}+100=P\left(1+\frac{10}{100}\right)^3$$

$$33.09+100=P\times\frac{11\times11\times11}{10\times10\times10}$$

$$\frac{133.09\times10\times10\times10}{11\times11\times11}=P$$

$$\frac{133090}{1331}=P$$

= ₹ 99.9925 ≅ ₹100

143. Mr. X invested ₹300 for 3 years at a simple interest rate of 11.03%. How much should Mr. Y invest after 3 years to get the same amount to return, but at a compound interest rate of 10%?

(a) ₹ 250 (b) ₹ 300

(c) ₹ 350 (d) ₹ 400

RRB Group-D – 23/10/2018 (Shift-I)

Ans. (b)

The amount given on compound interest is ₹P_1.

As per the question-

$$P_1\left(1+\frac{r}{100}\right)^n=\frac{P_2\times R\times t}{100}+P_2$$

$$P_1\left(1+\frac{10}{100}\right)^3=\frac{300\times3\times11.03}{100}+300$$

$$P_1\left(\frac{11}{10}\right)^3=99.27+300$$

$$P_1\left(\frac{11}{10}\times\frac{11}{10}\times\frac{11}{10}\right)=399.27$$

$$P_1=\frac{39927\times10}{1331}=299.97\approx₹300$$

144. Raju invested ₹5000 on annual compound interest rate for 3 years After 3 years he received the ₹1655 more, then what is the rate of interest.

(a) 40 (b) 30

(c) 10 (d) 20

RRB Group-D – 11/10/2018 (Shift-III)

Ans : (c) Principal = ₹ 5000

Compound interest = ₹ 1655

Compound amount = 5000 + 1655

= ₹ 6655

Time = 3 Year, Rate = R %

Formula–

$$\text{Compound amount} = \text{Principal}\left[1+\frac{\text{Rate}}{100}\right]^{\text{Time}}$$

$$6655 = 5000\left(1+\frac{R}{100}\right)^3$$

$$\frac{6655}{5000} = \left(1+\frac{R}{100}\right)^3$$

$$= \frac{1331}{1000} = \left(1+\frac{R}{100}\right)^3$$

$$= \left(\frac{11}{10}\right)^3 = \left(1+\frac{R}{100}\right)^3$$

$$\Rightarrow \frac{11}{10} - 1 = \frac{R}{100}, \Rightarrow \frac{1}{10} = \frac{R}{100}$$

R = 10%

145. A loan of ₹ 305 was taken at a certain rate per annum of compound interest, for 3 years After three years the amount repaid was ₹ 670, calculate the rate of compound interest.

(a) 30% (b) 35%

(c) 33% (d) 25%

RRB NTPC 27.04.2016 Shift : 1

Ans : (a) Amount $(A) = P\left(1+\frac{r}{100}\right)^n$

Time = 3Year

$$670 = 305\left(1+\frac{r}{100}\right)^3$$

$$\frac{670}{305} = \left(1+\frac{r}{100}\right)^3$$

$$2.197 = \left(1+\frac{r}{100}\right)^3$$

$$1+\frac{r}{100} = 1.3$$

$$\frac{r}{100} = 0.3$$

r = 30% Annual

Type - 6

146. In what time will ₹4400 become ₹4576 at 8% per annum compound interest rate. The interest compounded half-yearly?

(a) 6 month

(b) 2 Year

(c) 7 month

(d) 1 Year

RRB ALP & Tec. (17-08-18 Shift-I)

Ans : (a) Let, Time = n year = 2n half year

Given,

Amount (A) = ₹4576

Principal (P) = ₹4400

Rate $(r) = \frac{8\%}{2}$

= 4% Half yearly

As per the question,

Time = 2n Half year

$A = P\left(1+\frac{r}{100}\right)^n$ from,

$$\Rightarrow 4576 = 4400\left(1+\frac{4}{100}\right)^{2n}$$

$$\frac{4576}{4400} = \left(1+\frac{4}{100}\right)^{2n}$$

$$\left(\frac{26}{25}\right)^1 = \left(\frac{26}{25}\right)^{2n}$$

$$\Rightarrow \quad 2n = 1 \Rightarrow n = \frac{1}{2}$$

So required time = 6 months

147. Under a new scheme, a bank offers an interest of 30% per annum compounded annually. Suraj deposits ₹ 10,000 under this new scheme and at the end of the tenure receives ₹ 28,561. What was the tenure of the scheme that Suraj had chosen?

(a) 2 Year (b) 3.5 Year

(c) 4 Year (d) 4.5 Year

RRB NTPC 28.04.2016 Shift : 2

Ans : (c) A = ₹28,561

P = ₹10,000

r = 30%

n = ?

$$\because A = P\left(1+\frac{r}{100}\right)^n$$

$$28561 = 10000\left(1+\frac{30}{100}\right)^n$$

$$\frac{28561}{10000} = \left(\frac{130}{100}\right)^n = \left(\frac{13}{10}\right)^n = \left(\frac{13}{10}\right)^4$$

$\therefore$ n = 4 Year